

Ext

5

In this chapter, we will learn about

- the relationship between gas pressure, temperature and volume
- the general gas law
- the kinetic theory of gases

Gases

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5.1 The gas laws

It is hard to burst a balloon by simply squeezing it unless a large force is applied (Fig 5.1a). However, by using a sharp pin, you can easily burst the balloon with very little force (Fig 5.1b). Do you know the physics behind this?

Flipped classroom
Watch a video and answer the questions.



Fig 5.1a *It is difficult to burst a balloon with bare hands.*



Fig 5.1b *Only a small force is needed to burst a balloon with a pin.*

1 Pressure

Cross-link
Force is introduced in Book 2 Ch 3. Its unit is newton (N).

The above situations can be explained in terms of **pressure**. Pressure, denoted by p, is defined as the normal (perpendicular) force exerted on a surface per unit area.

$$\text{Pressure} = \frac{\text{normal force on a surface}}{\text{area of the surface}}$$

Ts may remind Ss that force is a vector while pressure is a scalar.

pressure (Pa) $p = \dfrac{F}{A}$ force on surface (N); area of surface (m^2)

The unit of pressure is **pascal** (**Pa**).

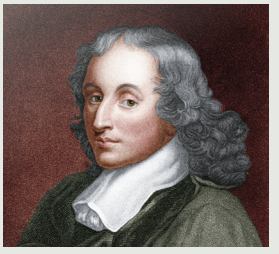
$$1 \text{ Pa} = 1 \text{ N m}^{-2}$$

Know more

Blaise Pascal (1623–1662)

Blaise Pascal was a French scientist. He contributed greatly to the study of fluids and clarified the concepts of pressure and vacuums. The unit of pressure was named after him.

OUP eResources



pressure 壓強　　pascal (Pa) 帕斯卡　　Blaise Pascal 帕斯卡

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Example 1 | Pressure exerted on a balloon

Two identical balloons are manipulated in two ways. One is squeezed by hand and the other is punctured with a pin. Assume the forces acting on each balloon are perpendicular to the balloon's surface.

(a) If the area of balloon that is in contact with the hands is 1.6×10^{-2} m^2, estimate the pressure exerted on the balloon by the hands when squeezing it with a total force of 80 N.

(b) If the pin exerts a force of 6 N on the balloon and the area of the pin's tip is 4×10^{-6} m^2, estimate the pressure exerted on the balloon by the pin.

Solution

(a) $$\text{Pressure} = \frac{F}{A}$$
$$= \frac{80}{1.6 \times 10^{-2}}$$
$$= 5000 \text{ Pa}$$

(b) $$\text{Pressure} = \frac{F}{A}$$
$$= \frac{6}{4 \times 10^{-6}}$$
$$= 1.5 \times 10^{6} \text{ Pa}$$

This shows that the pin exerts a greater pressure on the balloon even with a much smaller force. Therefore, it is easier to burst a balloon with a pin.

Know more

Nail Bed and Snowshoes

Do you think standing on a board of nails would be dangerous? Actually a person may stand on a board of nails without getting hurt. The pressure exerted by the large number of nails on the feet may not be very high due to a large total contact area.

The same concept can be applied to help us walk on thick soft snow. Our feet may sink into the snow when we wear ordinary shoes. Snowshoes have a much larger base area so that the pressure exerted by them on the snow is much smaller. This prevents us from sinking into the snow.



Standing on a board of nails.



Walking on snow with snowshoes.

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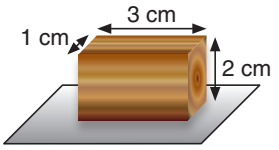
Google Forms Microsoft Forms

Quick check 1

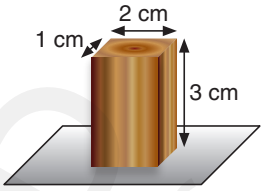
→ E-version in OUP Exercise Platform is also available.

1 A block is placed on the floor in two different orientations.

Case I



Case II



In which case is the pressure exerted on the floor by the block higher?

A Case I

B Case II

C The pressures are the same in both cases.

2 Complete the table by applying $p = \frac{F}{A}$.

	p / Pa	F / N	A / m^2
(a)		120	0.05
(b)	8.5×10^5		0.24

DSE goal

Realize the existence of gas pressure.

DSE exam

20(1B)Q2(a)(i), (ii)

2 Gas pressure

a Atmospheric pressure

Consider a sealed plastic bag filled with air. When we press the bag, we can feel a resistive force opposing our push (Fig 5.1c). The resisting force actually comes from the gas pressure of the air inside the bag.

The pressure from the air in the atmosphere around us is called the **atmospheric pressure**. It acts on all surfaces exposed to the atmosphere, including our bodies (Fig 5.1d). We seldom notice it because our bodies have already adapted to it.

The average atmospheric pressure at sea level is set to 1 atm, which is approximately equal to 100 kPa (also known as 1 bar). The pressure of the atmosphere is lower at higher altitudes since the air is thinner. Some mobile devices have a built-in pressure sensor. They can be used to measure the atmospheric pressure with a suitable app.



Fig 5.1c Gas pressure is exerted inside the bag.

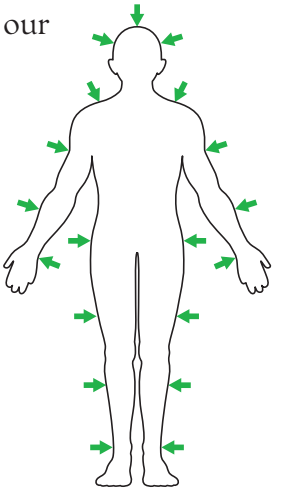


Fig 5.1d Atmospheric pressure acts on us all the time.

atmospheric pressure 大氣壓強

Ext

Atmospheric pressure is in action every day. If we keep drinking a carton of juice without letting air go into the carton, the carton will collapse in on itself. This is because the pressure inside the carton becomes lower than the atmospheric pressure outside (Fig 5.1e).

Similarly, when we release a suction cup after squeezing it, the gas pressure inside the cup becomes lower than the atmospheric pressure (Fig 5.1f). Since air is not allowed to flow into the enclosed cavity of the cup, a gas pressure difference between the inside and outside of the cup is sustained. The higher atmospheric pressure outside keeps pressing on the cup, making it attach firmly to the wall.



Fig 5.1e *The carton collapses in on itself.*

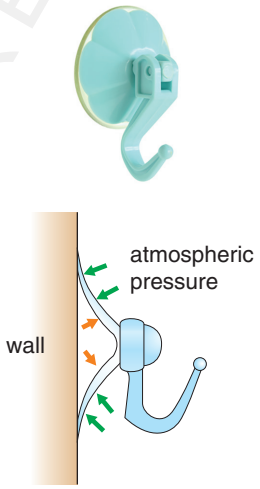


Fig 5.1f *Suction cup.*

Know more

Atmospheric pressure at different altitudes

If you seal an empty bottle with its cap at the top of a high mountain and bring the bottle down to the foot of the mountain, the bottle will gradually collapse. Figure a shows an example.

The atmospheric pressure increases as altitude decreases (Fig b). When the bottle is brought down from the top of the mountain, the atmospheric pressure becomes higher than the gas pressure inside the bottle and therefore the bottle collapses.

going down from the top of Mauna Kea volcano in Hawaii



Fig a

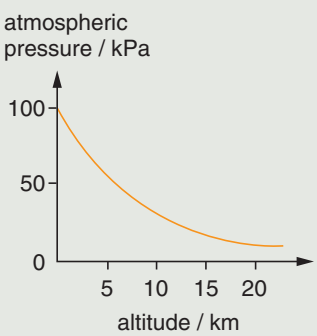
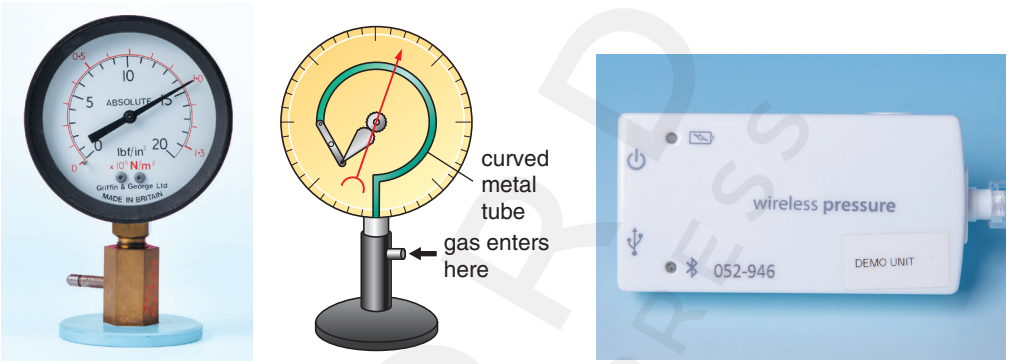


Fig b

Ext

b Measuring gas pressure

Gas pressure can be measured using a **Bourdon gauge** or a pressure sensor (Fig 5.1g). A Bourdon gauge has a curved metal tube that coils when gas pressure is applied. This moves the pointer around a dial.



(i) A Bourdon gauge and its structure.

(ii) A wireless pressure sensor.

Fig 5.1g *Apparatuses for measuring gas pressure.*

3 Boyle's law

DSE goal
Verify Boyle's law.

DSE exam
12(1B)Q2(a), 14(1B)Q2(a), 15(1B)Q2(a), 18(1A)Q4

In the 17th century, scientists began to study the physical properties of gases. *Robert Boyle* found the relationship between the pressure and the volume of a fixed mass of gas at constant temperature.

Experiment 5a Boyle's law (pressure–volume relationship of gas)

Simulation Video
→ This simulation is a 'virtual expt' on Boyle's law. It is available on OUP web in PC version (.exe) only.

→ This video demonstrates Expt 5a.

→ This is a Video Quiz about this experiment.

1 Set up the apparatus as shown (Fig a). Slowly apply the foot-pump to increase the pressure p of the air column.

2 Take several sets of p and V (volume of air column). Plot a p–V graph and a p–$\frac{1}{V}$ graph.

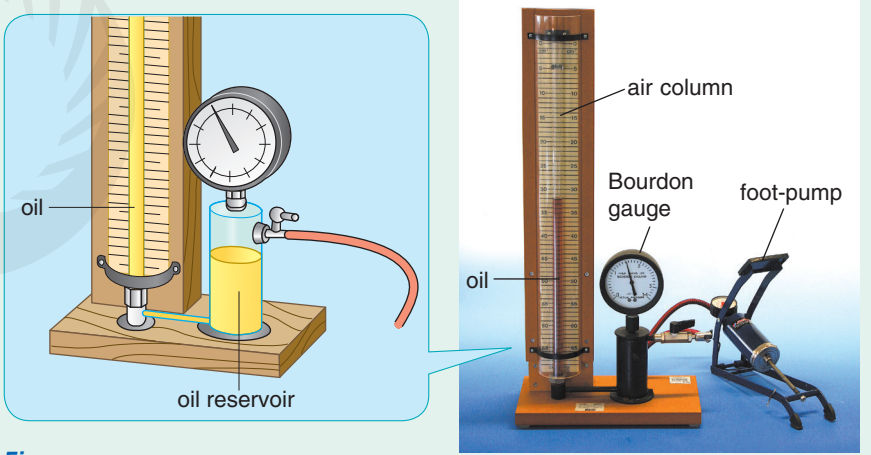


Fig a

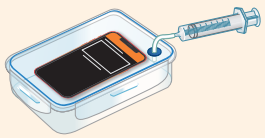
Bourdon gauge 布爾登氣壓計　　Robert Boyle 波義耳

Ext

Phone experiment

Boyle's law

You may also measure the air pressure with your smartphone (with a built-in pressure sensor) to verify Boyle's law.



▶ p.191

3 Alternatively, connect a pressure sensor to a syringe (Fig b). Push or pull the piston of the syringe and take several sets of p and V. Plot a p–V graph and a p–$\frac{1}{V}$ graph.

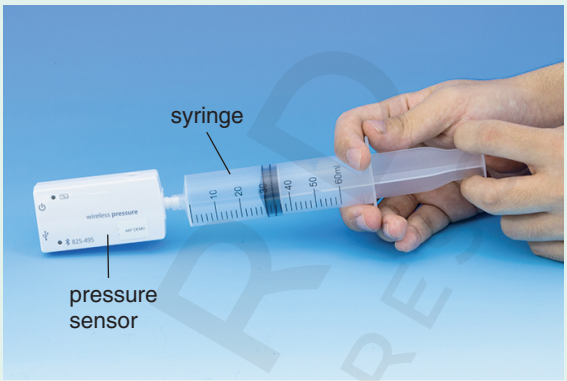


Fig b

Precaution

1 Change the volume of air slowly to minimize any change in air temperature.

2 Hold the syringe with the least contact area possible in step 3 to minimize any change in air temperature due to the heat transfer through the contact.

3 If a rubber tubing is needed to connect the syringe and the sensor, it should be as short as possible to minimize the errors in measuring the volume of air. The reading on the syringe does not include the air in the tubing.

Discussion

What is the relationship between the pressure (p) and the volume (V) of a gas?

In Experiment 5a, the graph of pressure against volume gives a curve (Fig 5.1h). However, when the pressure is plotted against $\frac{1}{\text{volume}}$, a straight line passing through the origin is obtained (Fig 5.1i).

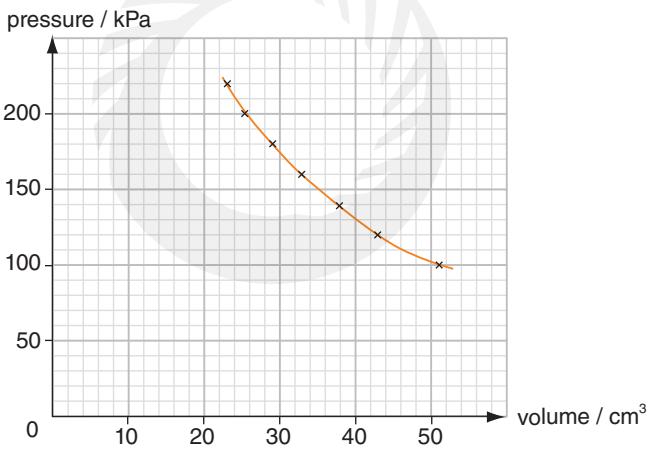


Fig 5.1h *Graph of pressure against volume.*

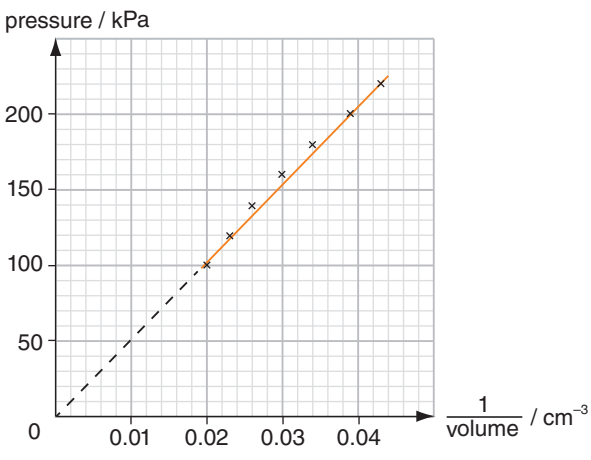


Fig 5.1i *Graph of pressure against* $\frac{1}{\text{volume}}$.

Ext

Know more

Robert Boyle (1627–1691)



Robert Boyle was an Irish scientist who made important contributions to physics and chemistry. Besides the formulation of Boyle's law, he is famous for being the first scientist to keep accurate records of experimental results.

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The straight line graph shows that the pressure of a gas is inversely proportional to its volume (at constant temperature). This relationship is called **Boyle's law**:

For a gas with a fixed mass and temperature, its pressure is inversely proportional to its volume, i.e.

$$p \propto \frac{1}{V} \quad \text{(for constant } T\text{)}$$

pressure of a gas with fixed mass and temperature (Pa)

volume of the gas (m^3)

This also means that pV = constant. If the values of pressure and volume are initially p_1 and V_1, and eventually change to p_2 and V_2, then

$$pV = \text{constant}$$

or

$$p_1V_1 = p_2V_2 \quad \text{(for constant } T\text{)}$$

Know more

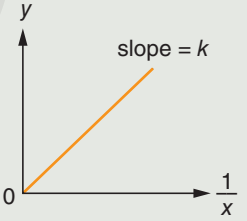
Inverse proportion $y \propto \frac{1}{x}$

When the graph of y plotted against $\frac{1}{x}$ is a straight line through the origin, we say that y is inversely proportional to x, i.e. $y \propto \frac{1}{x}$. In this case, we can write

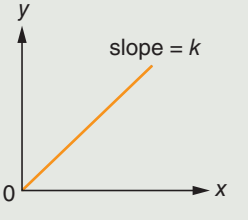
$$y = k \times \frac{1}{x} \text{ or } xy = k$$

where the constant k is the slope of the straight line.

Note the difference between inverse proportion and direct proportion. y is directly proportional to x ($y \propto x$) when the graph of y plotted against x is a straight line through the origin.



Graph showing $y \propto \frac{1}{x}$.



Graph showing $y \propto x$.

Boyle's law 波義耳定律

Ext

Example 2 | Air toy gun

Initially, the body of an air toy gun contains 250 cm^3 of air with an atmospheric pressure of 100 kPa. The soft bullet is installed on the outlet of the gun. When the body is pressed, the volume of the body decreases. The bullet will be expelled when the pressure of the air reaches 125 kPa. Find the volume of the air in the body when the bullet is expelled.



Solution

By Boyle's law,

$$p_1V_1 = p_2V_2$$

$$100 \times 250 = 125 \times V_2$$

$$V_2 = 200 \text{ cm}^3$$

100 kPa, 250 cm^3 → 125 kPa, V_2 = ?

The volume of the air in the body is 200 cm^3.

Tip

Express the same quantity in the same unit (p_1 and p_2 in kPa, and V_1 and V_2 in cm^2 in this case).

Learn by practice 1

1 A sealed syringe is filled with 50 cm^3 of air with an atmospheric pressure of 100 kPa. If the volume inside the syringe is reduced to 45 cm^3, what will be the new pressure? Assume the temperature of the air remains unchanged.

2 The volume of the air inside a sealed bag of potato chips is 50 cm^3 at 0.994 atm, i.e. 99.4 kPa (Fig a). Find the volume of the air inside the bag if the pressure becomes 79.0 kPa. Assume the temperature of the air inside the bag remains unchanged.



Fig a

Ext

DSE goal

Determine pressure–temperature relationship of a gas.

Determine absolute zero by the extrapolation of pressure–temperature relationship.

Use kelvin as a unit of temperature.

DSE exam

21(1B)Q2(b)(i), 23(1A)Q3

- Do not heat the flask in an oven with an exposed heating coil. Such a heating coil may ignite the cotton.
- The flask is very hot. Do not touch it.

Simulation Video

→ This simulation is a 'virtual expt' demonstrating the pressure–temperature relationship of a gas. It is available on OUP web in PC version (.exe) only.

→ This video demonstrates Expt 5b.

→ Phone experiment worksheet for 'How gas pressure changes with temperature' is available on OUP web.

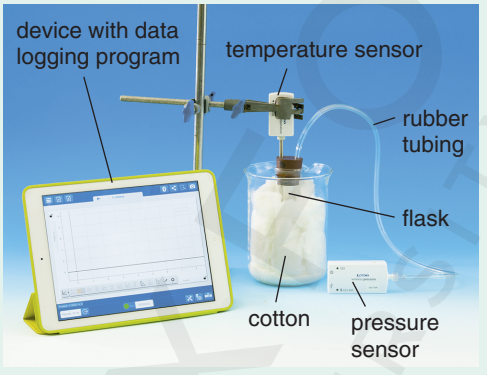
4 Pressure–temperature relationship

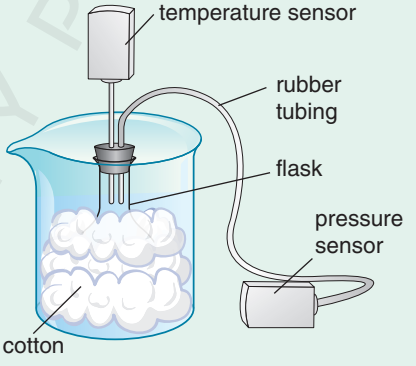
What will happen to the pressure of a gas if the temperature changes while the volume and the mass of the gas are kept constant? Let us find the answer in the following activity and experiment.

Experiment 5b / Pressure–temperature relationship of gas

1 Heat a flask wrapped with cotton in an oven set at 150 °C for about 20 minutes. Then take out the flask and seal it.

2 Measure the temperature T and pressure p of the gas inside the flask. Plot a graph of p against T.





Discussion

1 What is the relationship between the pressure (p) and the temperature (T) of a gas?

2 Is it necessary to keep the rubber tubing short?

A straight line graph is obtained in Experiment 5b (Fig 5.1j). Therefore, for a gas with a fixed mass and volume, there is a **linear relationship between its pressure and temperature**.

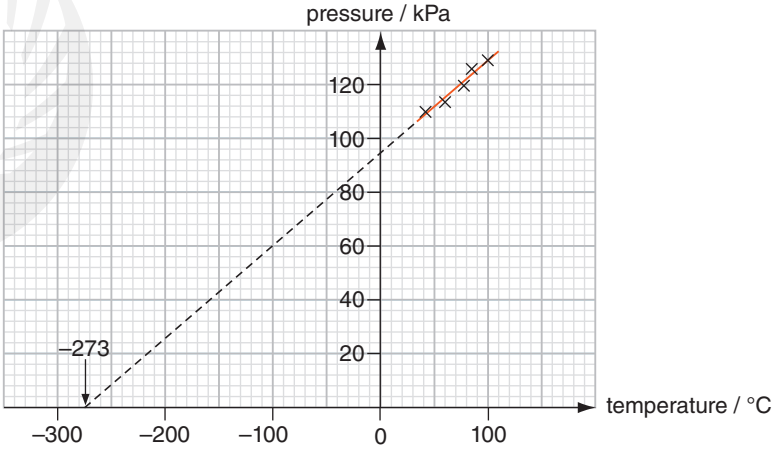


Fig 5.1j Graph of pressure against temperature.

Ext

a The Kelvin temperature scale

More precisely, the value is −273.15 °C.

If the graph in Figure 5.1k is extrapolated to the temperature axis, it will cut the axis at a temperature of around −273 °C. Experiments using different gases give the same intercept at the temperature axis.

If the pressure axis is shifted leftwards to the intercept (−273 °C), the prolonged graph line will pass through the origin.

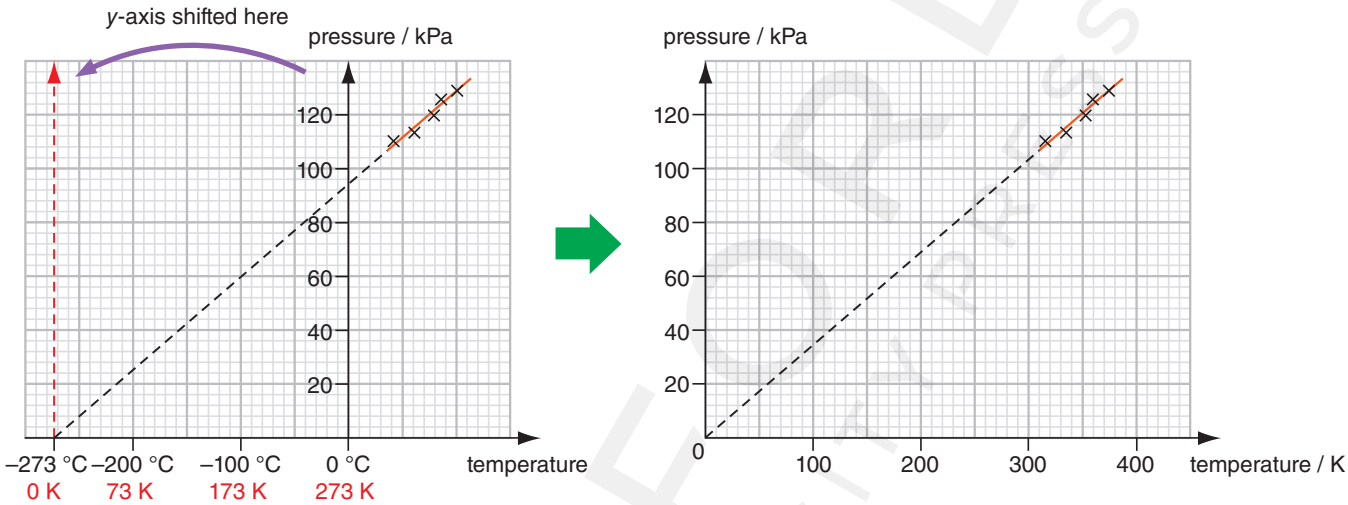


Fig 5.1k *Graph of pressure against temperature, with the pressure axis shifted to the left by 273 °C.*

Ts may point out that a straight-line graph implies a linear relationship. However, the straight line must pass through the origin for a proportional relationship.

Therefore, a direct proportional relationship of pressure and temperature can be obtained by defining a new temperature scale called the **Kelvin temperature scale**.

$$T_K = T_C + 273$$

Kelvin temperature (K) — Celsius temperature (°C)

Caution
Note that the symbol is K, not °K.

The unit for the Kelvin temperature scale is **kelvin**. Its symbol is **K**.

The temperature of zero kelvin cannot be reached in practice although we can get down to millionths of a kelvin.

In Chapter 2, we learned that absolute zero is the lowest temperature an object can reach. The average kinetic energy of the molecules reaches the minimum value at this temperature. In fact, zero kelvin is the absolute zero. Therefore, the Kelvin temperature scale is also called the **absolute temperature scale**.

We can now give temperatures in kelvins. For example, the freezing and boiling points of water are 273 K and 373 K respectively (Fig 5.1l). Note that a temperature change of 1 K is equal to a temperature change of 1 °C.

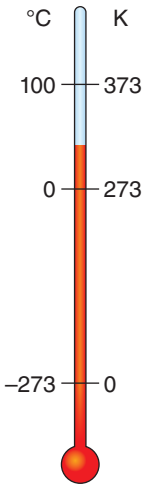


Fig 5.1l *Celsius temperature and Kelvin temperature.*

Kelvin temperature scale 開氏溫標 kelvin (K) 開 absolute temperature scale 絕對溫標

Ext

Example 3 Investigating *p*–*T* relationship of a gas

Linda sets up the following apparatus to study the relationship between pressure p and temperature T of a fixed mass of gas at a constant volume (Fig a). The flask is heated by a water bath. The pressure of the air in the flask is measured by a Bourdon gauge at various temperatures. The result is plotted as a straight line graph as shown in Figure b.

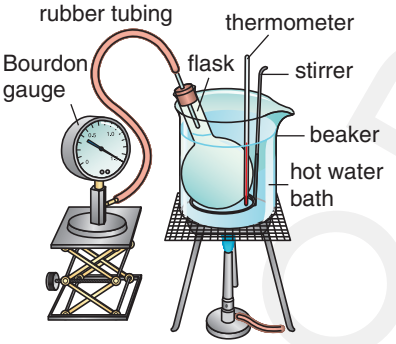


Fig a

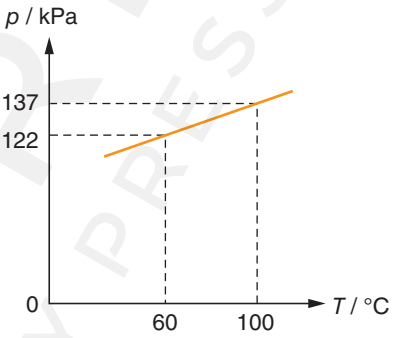


Fig b

(a) Estimate the absolute zero from the graph.

(b) Suggest what Linda should do to ensure the air in the flask reaches the same temperature as the water bath before the readings are taken.

(c) Linda believes that shortening the rubber tubing of the Bourdon gauge could improve the accuracy of the result. Is she correct? Explain briefly.

(d) Point out two more modifications that can improve the accuracy of the result.

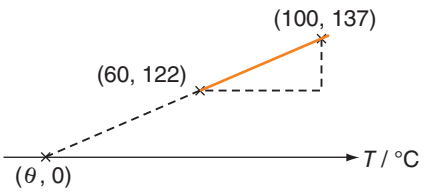
Solution

(a) Let θ °C be the absolute zero.

$$\frac{122 - 0}{60 - \theta} = \frac{137 - 122}{100 - 60}$$

$$\theta = -265$$

The absolute zero is −265 °C.



(b) She should stop heating the water bath and stir the water well. Then wait until the temperature is steady to take the readings.

(c) She is correct. Since the rubber tubing is not in direct contact with the water bath, the temperature of the air inside the tubing increases more slowly than that inside the flask and is probably lower than the readings taken. Shortening the rubber tubing can reduce the error due to this.

(d) The whole flask, including the neck, should be immersed in the water bath.

The thermometer should not touch the bottom of the beaker.

Ext

Google Forms

Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) The unit for the Kelvin temperature scale is °K. (T/F)

(b) A temperature difference of one kelvin is the same as that of one degree Celsius. (T/F)

2 Complete Table a.

	Temperature (in degree Celsius)	Temperature (in kelvin)
Body temperature	37	
Melting point of mercury		234
Absolute zero		

Table a

b Pressure law

Using the Kelvin temperature scale, the p–T graph is a straight line passing through the origin, indicating a direct proportional relationship of pressure and temperature (at constant volume). This relationship is called **pressure law**:

Pressure law is also known as pressure–temperature relationship.

For a gas with a fixed mass and volume, its pressure is directly proportional to its Kelvin temperature, i.e.

$$p \propto T \quad \text{(for constant } V\text{)}$$

pressure of a gas with fixed mass and volume (Pa)

temperature of the gas (K)

This also means that the ratio of the pressure to the Kelvin temperature is a constant, i.e.

$$\frac{p}{T} = \text{constant}$$

or

$$\frac{p_1}{T_1} = \frac{p_2}{T_2} \quad \text{(for constant } V\text{)}$$

pressure law 氣壓定律

Ext

Example 4 Pressure inside an air-tight jar

An air-tight jar contains a gas with a pressure of 100 kPa at 20 °C. If the jar is put in a refrigerator so that the temperature of the gas drops to 4 °C, what will be the gas pressure inside the jar?



Solution

By pressure law,

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

100 kPa, 20 °C → p_2 = ?, 4 °C

$$\frac{100}{20 + 273} = \frac{p_2}{4 + 273}$$

$$p_2 = 94.5 \text{ kPa}$$

The gas pressure inside the jar is 94.5 kPa.

Tip

Remember to convert °C to K and express p_1 and p_2 in the same unit (kPa in this case).

As the jar is rigid, it will not change its shape even if the internal and external pressures are different. However, if the difference is extremely large, the jar will burst.

Watch a video showing a heated metal can suddenly cooled by cold water. The can collapses dramatically.



Video

→ This video shows the 'collapsing can" expt. The can is heated and the air inside is pushed out of the can. When the can is then cooled by cold water with its opening blocked by the water, the can collapses.

Learn by practice 2

1 A sealed container holds a gas at 300 K. If the gas is heated so that its pressure increases from 60 kPa to 180 kPa, what is its final temperature? Assume the volume of the gas is fixed.

2 A sealed container is filled with air at a temperature of 20 °C and a pressure of 9×10^4 Pa. Find the pressure of the air if its temperature increases to 40 °C. Assume the volume of the air is fixed.

DSE goal

Determine volume–temperature relationship of a gas.

Determine absolute zero by the extrapolation of volume–temperature relationship.

DSE exam

13(1B)Q2(b)

5 Volume–temperature relationship

In the following experiment, we will study how the volume of a gas changes with temperature while the gas pressure is kept constant.

Ext

→ This simulation is a 'virtual expt' demonstrating the volume–temperature relationship of a gas. It is available on OUP web in PC version (.exe) only.

→ This video demonstrates Expt 5c.

Simulation
Video

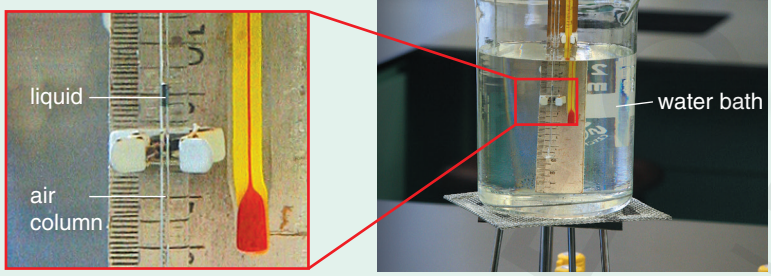
Experiment 5c / Volume–temperature relationship of gas

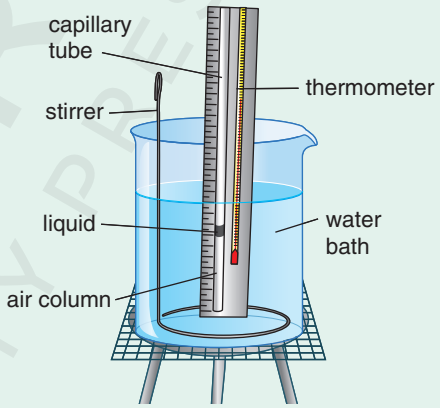
1 Set up the apparatus as shown. Immerse the capillary tube in the water bath. Heat the water bath. Record the lengths L of the air column at different temperatures T.

2 Plot a graph of L (which represents the volume of the air) against T.

The heated set-up is very hot. Do not touch it.

An air column is trapped inside the capillary tube by a drop of liquid. The capillary tube is open at the top, so the air column is at the same pressure throughout the experiment.





Precaution

Before taking a reading, stop heating and stir well to ensure the temperature is the same throughout the water.

Discussion

What is the relationship between the volume and the temperature of a gas?

→ This is a Video Quiz about this experiment.

The graph obtained in Experiment 5c is a straight line (Fig 5.1m). Therefore, for a gas with a fixed mass and constant pressure, there is a **linear relationship between its volume and temperature**.

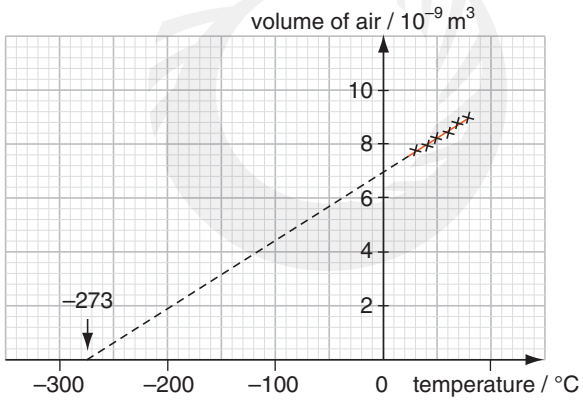


Fig 5.1m *Graph of volume against temperature.*

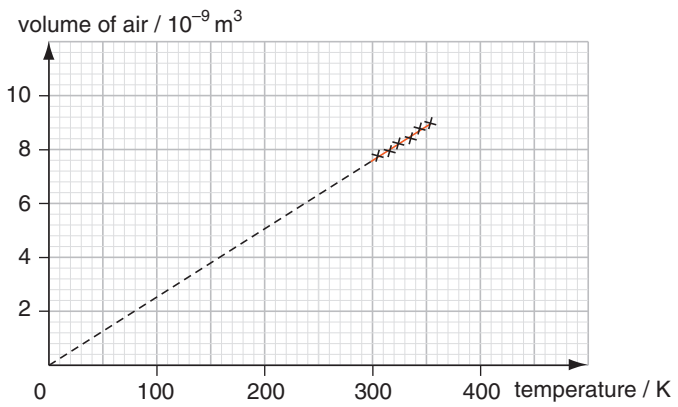


Fig 5.1n *Graph of volume against temperature, with the volume axis shifted to the left by 273 °C.*

Ext

The V–T graph also cuts the temperature axis at around –273 °C when extrapolated. Experiments with different gases give the same intercept at the temperature axis.

Similar to the p–T graph, if the Kelvin temperature scale is used, the V–T graph is a straight line passing through the origin (Fig 5.1n on p.160), indicating a direct proportional relationship of volume and temperature (at constant pressure). This relationship is called **Charles' law**:

Charles' law is also known as volume–temperature relationship.

For a gas with a fixed mass and pressure, its volume is directly proportional to its Kelvin temperature, i.e.

$$V \propto T \quad \text{(for constant } p\text{)}$$

volume of a gas with fixed mass and pressure (m^3)

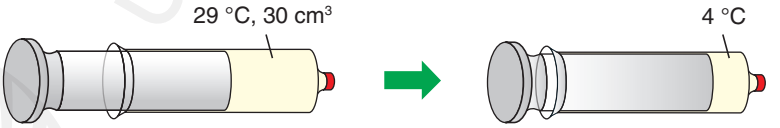
temperature of the gas (K)

This also means that the ratio of the volume to the Kelvin temperature is a constant, i.e.

$$\frac{V}{T} = \text{constant} \quad \text{or} \quad \frac{V_1}{T_1} = \frac{V_2}{T_2} \quad \text{(for constant } p\text{)}$$

Example 5 Volume of air in a syringe

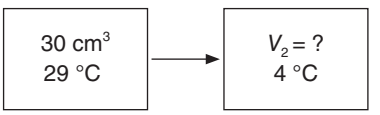
A sealed syringe is filled with 30 cm^3 of air at a temperature of 29 °C. The syringe is cooled to 4 °C in a freezer. What is the new volume of the air? Assume the air pressure inside the syringe remains constant.



Solution

By Charles' law,

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$



Tip

Remember to convert the temperature from unit of °C to K.

$$\frac{30}{29 + 273} = \frac{V_2}{4 + 273}$$

$$V_2 = 27.5 \text{ cm}^3$$

The new volume of the air is 27.5 cm^3.

Charles' law 查理定律

Ext

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 Which of the following phenomena can be explained by Charles' law?

A



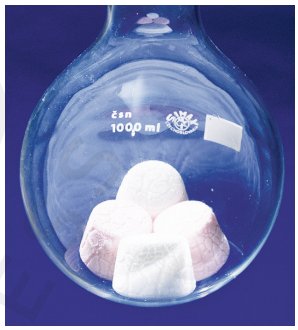
Balloons shrink when cooled.

B



Bubbles pop when bubble wrap is squeezed.

C



Marshmallows inside a sealed flask inflate when air is pumped out from the flask.

2 Match each of the following laws with the correct relation(s) and condition(s).

(a) Charles' law (b) Boyle's law (c) Pressure law

Relation	
(i)	$p_1V_1 = p_2V_2$
(ii)	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$
(iii)	$\frac{p_1}{T_1} = \frac{p_2}{T_2}$

Condition	
(I)	constant volume
(II)	constant temperature
(III)	fixed mass
(IV)	constant pressure

3 *True or false:*

(a) For a gas with a fixed mass and pressure, its V–T graph is a straight line which cuts the temperature axis at 0 °C. (T/F)

(b) In applying $\frac{p_1}{T_1} = \frac{p_2}{T_2}$, both temperatures must be expressed in the same unit, either in kelvins or degrees Celsius. (T/F)

Learn by practice 3

1 Some oxygen has a volume of 2.5 m^3 at 300 K. If its pressure and mass remain constant, what is its volume at 390 K?

2 The pressure of a fixed mass of gas is kept constant. Initially, the gas has a volume of 15.8 cm^3 and a temperature of 0 °C. Under what temperature would the gas expand to 20.4 cm^3?

Ext

DSE goal

Combine the three relationships (p–V, p–T and V–T) of a gas to obtain the relationship $\frac{pV}{T}$ = constant.

Apply the general gas law $pV = nRT$ to solve problems.

Realize the condition that at high temperature and low pressure a real gas behaves as an ideal gas.

DSE exam

12(1A)Q3, 13(1A)Q4, 16(1B)Q2, 19(1B)Q2, 20(1A)Q3, 21(1B)Q2(c), 22(1A)Q3, etc.

6 The general gas law and ideal gases

a The general gas law

From the previous parts, we know the following relationships:

Boyle's law: $p \propto \frac{1}{V}$ (for constant T)

Pressure law: $p \propto T$ (for constant V)

Charles' law: $V \propto T$ (for constant p)

These relationships can be combined into a single expression:

$$pV \propto T$$

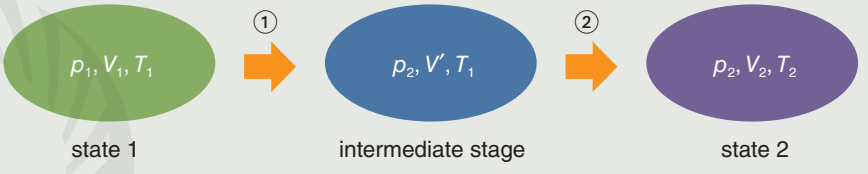
This also means that, for a fixed mass of gas, the ratio of the product of the pressure and the volume to the Kelvin temperature is a constant, i.e.

$$\frac{pV}{T} = \text{constant} \quad \text{or} \quad \frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$$

Know more

Combining three proportional relationships into one

The three relationships can be combined into one expression in the following way. Consider a fixed mass of gas changes from state 1 (with p_1, V_1 and T_1) to state 2 (with p_2, V_2 and T_2) via an intermediate stage.



① By keeping T_1 constant and changing p_1 to p_2, Boyle's law can be applied.

$p_1V_1 = p_2V'$

② By keeping p_2 constant and changing T_1 to T_2, Charles' law can be applied.

$\frac{V'}{T_1} = \frac{V_2}{T_2}$

Then, by eliminating V', the expression becomes:

$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$

Ext

Avogadro's number is named after the Italian physicist Amedeo Avogadro (1776–1856). In 1811, Avogadro proposed that equal volumes of all gases at the same temperature and pressure contain equal numbers of molecules. After this proposal, scientists tried hard to determine this number. In 1960, Avogadro's number was defined as the number of atoms in 12 g of carbon-12, which was found experimentally to be about 6.02×10^{23}.

Now we know that $pV \propto T$. Can the exact relationship between the three physical quantities be further known? This can be done by considering the molecular motion of a gas.

For a particular gas with constant pressure and temperature, the volume V of the gas is directly proportional to the number of gas molecules N, i.e. $V \propto N$.

Since the total number of molecules is usually very large, it is more convenient to use the number of **moles** (**mol**) instead. One mole contains 6.02×10^{23} molecules. This number is known as the **Avogadro's number** (N_A). Then, for n mol of gas, we have

$$N = nN_A \Rightarrow N \propto n$$

Combining this with the expression $pV \propto T$ and $V \propto N$:

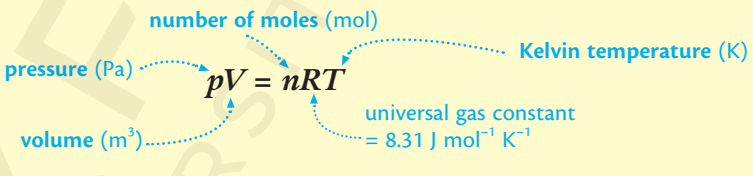
$$pV \propto T \Rightarrow pV \propto NT \Rightarrow pV \propto nT$$

By inserting the proportionality constant, the above proportion becomes an equation called the **general gas law**:

Caution

Unlike Boyle's law, pressure law and Charles' law, the units of the quantities in the general gas law are fixed.

Always convert the unit of temperature to kelvin when applying the general gas law.



The proportionality constant is the **universal gas constant** R. Its value is found through experiments and equals 8.31 J mol^{-1} K^{-1} for all gases.

If we use N instead of n, the general gas law will become

$pV = NkT$

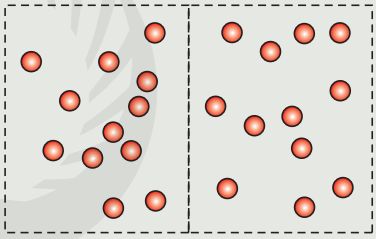
where k is called Boltzmann constant. It can be expressed as $\frac{R}{N_A}$ and its value is 1.38×10^{-23} J K^{-1}.

Know more

Number of molecules and number of moles

N and n are different physical quantities for counting the number of molecules, while N_A is simply a conversion factor between the two quantities. This is analogous to counting the number of marbles using different quantities:

Counting marbles



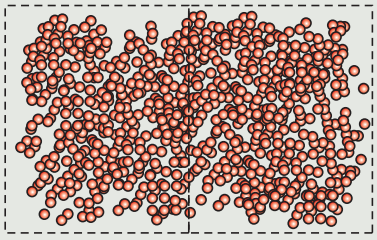
There are 24 marbles. ($N = 24$)

or

There are 2 dozen marbles. ($n = 2$)

1 dozen has 12 marbles.

Counting molecules



There are 12.04×10^{23} molecules. ($N = 12.04 \times 10^{23}$)

or

There are 2 moles of molecules. ($n = 2$)

1 mole has $N_A = 6.02 \times 10^{23}$ molecules.

mole (mol) 摩爾　Avogadro's number 阿佛加德羅數　general gas law 普適氣體定律
universal gas constant 普適氣體常數

Ext

b Ideal gases

At temperatures below the boiling point, a gas will condense and become a liquid.

Therefore, the general gas law is also called the *ideal gas law*.

Unless otherwise specified, we shall assume all gases can be described by the general gas law in this book.

A gas that obeys the general gas law is called an **ideal gas**. A real gas does not follow the equation precisely, but it behaves like an ideal gas at **low pressures** and **high temperatures** (well above its boiling point). The general gas law becomes accurate enough in describing the behaviour of a real gas under these conditions.

Example 6 Using the general gas law to determine the change in pressure

Carbonated drinks are beverages that contain dissolved carbon dioxide. When a carbonated drink is shaken, the carbon dioxide molecules in the drink liquid escape into the space above the liquid and become carbon dioxide gas.



A sealed bottle of carbonated drink is shaken. Determine the change in gas pressure inside the bottle using the general gas law.

Solution

When the dissolved carbon dioxide in the liquid drink changes to gas, the number of moles of gas in the bottle increases ($n\uparrow$).

By $pV = nRT$, with the temperature and the volume of the gas in the bottle remaining constant (T and V constant), the gas pressure inside the bottle will increase ($\Rightarrow p\uparrow$).

The bottles used for carbonated drinks cannot withstand high gas pressure. A real-life accident occurred due to the build-up of the pressure in an air-tight carbonated drink bottle containing dry ice. For details, refer to the news website below.

The situation will be explained using the kinetic theory in Writing practice on p.171 of Ch 5.2.

Know more

Gases behaving like an idea gas at room temperature and atmospheric pressure

For gases to obey the general gas law, the pressure must be low enough such that the intermolecular forces become negligible. In fact, even at room temperature and atmospheric pressure, a lot of gases, such as H_2, He, O_2, N_2 and CO_2, can follow the relation described by the ideal gas law.

ideal gas 理想氣體　　ideal gas law 理想氣體定律

Ext

Example 7 Hot air balloon

A hot air balloon has a fixed volume of 400 m^3. Initially, the air is at a temperature of 27 °C and a pressure of 100 kPa. Take $R = 8.31$ J mol^{-1} K^{-1} and $N_A = 6.02 \times 10^{23}$ mol^{-1}.



(a) Find

(i) the number of moles of gas molecules, and

(ii) the number of gas molecules

inside the balloon.

(b) The number of gas molecules inside the balloon changes when the gas is heated. If the gas is now heated to 120 °C and the balloon rises to an altitude so that the gas pressure becomes 80 kPa, what is the number of moles of gas molecules inside the balloon?

Solution

(a) (i) By $pV = nRT$,

$$n = \frac{pV}{RT} = \frac{(100 \times 10^3)(400)}{(8.31)(27 + 273)} = 16\ 040 \text{ mol} \approx 16\ 000 \text{ mol}$$

There are 16 000 mol of gas molecules inside the balloon.

(ii) Number of gas molecules N

$= nN_A = 16\ 040 \times (6.02 \times 10^{23}) = 9.66 \times 10^{27}$

Tip
Express p in Pa.

(b) By $pV = nRT$,

$$\frac{nT}{p} = \frac{V}{R} = \text{constant}$$

16 040 mol 100 kPa 27 °C	→	n_2 = ? 80 kPa 120 °C

$$\frac{n_1T_1}{p_1} = \frac{n_2T_2}{p_2}$$

$$\frac{(16\ 040)(27 + 273)}{100} = \frac{n_2 \times (120 + 273)}{80}$$

$$n_2 = 9800 \text{ mol}$$

Tip
As this equation actually comes from the ratios of the quantities of the gas, you can express both p_1 and p_2 in kPa.

There are 9800 mol of gas molecules inside the balloon.

Ext

Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 Which of the following equations is/are correct? Put a tick in the corresponding box(es).

- ☐ Number of moles = N_A × number of molecules
- ☐ Number of molecules = N_A × number of moles
- ☐ $\frac{pV}{T}$ = constant (for constant n)
- ☐ $\frac{pT}{n}$ = constant (for constant V)
- ☐ $\frac{pV}{n}$ = constant (for constant T)
- ☐ $\frac{V}{nT}$ = constant (for constant p)
- ☐ $\frac{n}{p}$ = constant (for constant V and T)

2 Complete the table by applying $pV = nRT$. Given: R = 8.31 J mol^{-1} K^{-1}

	p	V	n	T
(a)	120 kPa	1500 cm^3		300 K
(b)		0.084 m^3	0.23 mol	50 °C

Learn by practice 4

1 The air in a bedroom is at 25 °C and with an atmospheric pressure of 100 kPa, and its volume is 77 m^3. Take R = 8.31 J mol^{-1} K^{-1}. Estimate

(a) the number of moles of air, and

(b) the number of air molecules

inside the bedroom.

2 A sealed syringe contains 80 cm^3 of air at a temperature of 27 °C and a pressure of 100 kPa. When the piston is quickly pushed inwards to reduce the volume to 70 cm^3, the temperature rises to 33 °C. What is the new pressure?

3 A bottle with its lid closed contains 0.025 mol of air inside. The temperature and pressure inside the bottle are 25 °C and 110 kPa respectively. The lid is opened for a while and is then closed again. The temperature of the air remains at 25 °C while the pressure drops to 100 kPa. Take R = 8.31 J mol^{-1} K^{-1}.

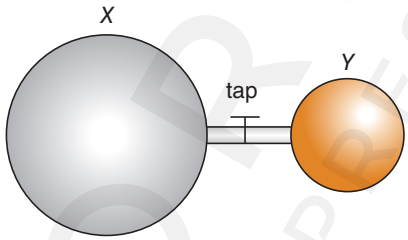
(a) Find the number of moles of air inside the bottle.

(b) Find the volume of the air inside the bottle.

Ext

Example 8 Mixing gases

Two containers X and Y are connected by a short narrow tube with a closed tap as shown below. Both of them contain the same ideal gas. The number of moles of the gas in X is n_X while that in Y is n_Y. The volume of X is three times that of Y while the pressure inside Y is three times that inside X. Assume the temperature of the gases is kept constant at T all the time.



(a) Find $\frac{n_X}{n_Y}$.

(b) The tap is opened to enable the gas to flow between the two containers.

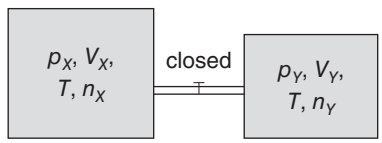
(i) Describe the direction of the gas flow. Explain your answer.

(ii) After some time, the net flow of gas between the containers stops. Find the percentage change of the pressure inside X.

Solution

(a) By $pV = nRT$, $\frac{pV}{n} = RT$

Since the gases in X and Y have the same temperature,



$$\frac{p_X V_X}{n_X} = \frac{p_Y V_Y}{n_Y}$$

$$\frac{n_X}{n_Y} = \frac{p_X}{p_Y} \times \frac{V_X}{V_Y} = \frac{1}{3} \times \frac{3}{1} = 1$$

Students may wrongly think that there would be no gas flow between the two containers since they contain the same amount of gas molecules.

Tip

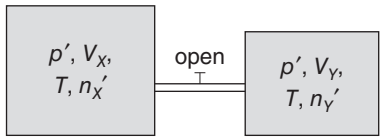
The total number of gas molecules in X and Y should be unchanged.

Tip

RT is common to all terms on both sides and can be cancelled out.

(b) (i) Since Y is at a higher pressure, gas will flow from Y to X.

(ii) The net flow of gas stops when the pressures in X and Y become the same.



$$n_X + n_Y = n_X' + n_Y'$$

$$\frac{p_X V_X}{RT} + \frac{p_Y V_Y}{RT} = \frac{p' V_X}{RT} + \frac{p' V_Y}{RT}$$

$$p_X(3V_Y) + (3p_X)V_Y = p'(3V_Y) + p'V_Y$$

$$6p_X = 4p'$$

$$p' = 1.5p_X$$

$$\text{Percentage change} = \frac{1.5p_X - p_X}{p_X} \times 100\% = +50\%$$

Ext

7 Graphs and gas laws

We have learned that the p–T graph is a straight line passing through the origin under constant volume V_1. But how is the p–T graph different from the original one when the same gas is kept at another constant volume V_2?

To answer this question, we should first rewrite the general gas law as $p = \left(\frac{nR}{V}\right)T$. It follows that the slope of the p–T graph is $\frac{nR}{V}$. Therefore, the slope will be larger if the gas is kept at a smaller constant volume V (Fig 5.1o). V–T graphs at different constant pressures can be drawn in a similar way (Fig 5.1p).

The equation of a straight line passing through the origin is in the form of $y = mx$. By comparing the two equations as follows, the slope of the p–T graph can be deduced.

$$p = \left(\frac{nR}{V}\right)T$$
$$\Updownarrow \quad \Updownarrow \quad \Updownarrow$$
$$y = \quad m \quad x$$

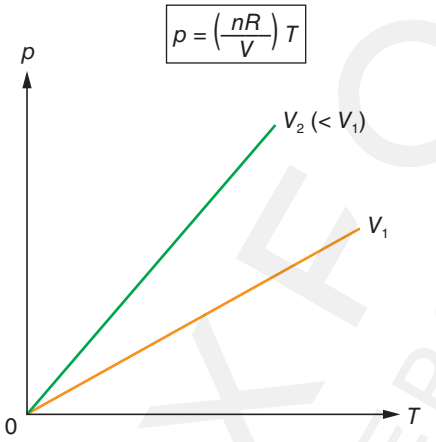


Fig 5.1o *The p–T graph with $V_2 < V_1$.*

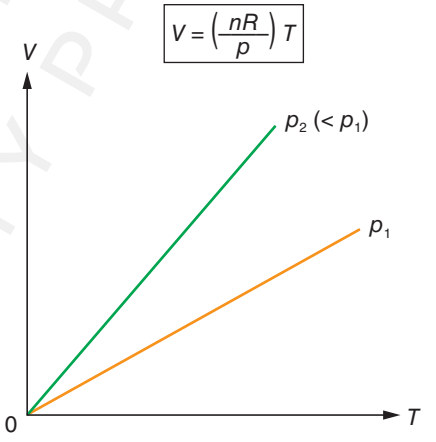
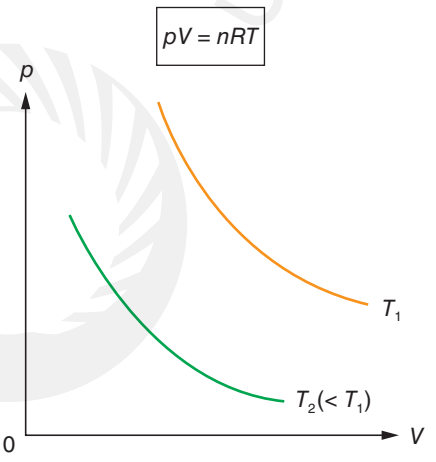
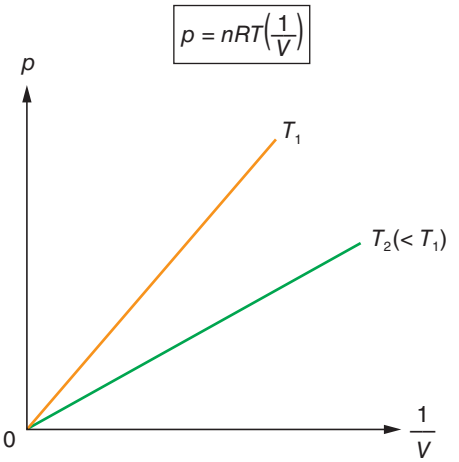


Fig 5.1p *The V–T graph with $p_2 < p_1$.*

By Boyle's law, if T_1 is larger than T_2, the p–V graph and p–$\frac{1}{V}$ graph will be as follows (Fig 5.1q).



(i) The p–V graph.



(ii) The p–$\frac{1}{V}$ graph.

Fig 5.1q *The graphs for Boyle's law with $T_2 < T_1$.*

Ext

Google Forms

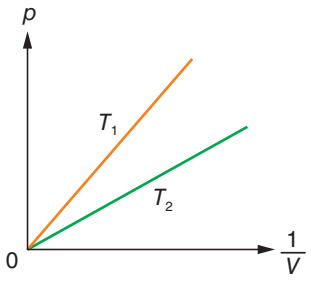
Microsoft Forms

Quick check 5

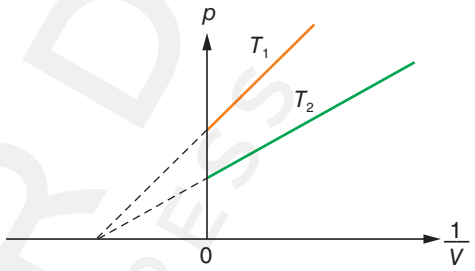
→ E-version in OUP Exercise Platform is also available.

1 Which of the following graphs correctly shows the relationship between the pressure p of a fixed mass of an ideal gas and its volume V at two different temperatures T_1 and T_2? It is given that $T_1 > T_2$.

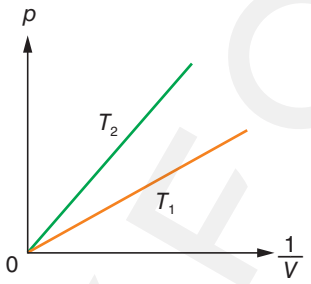
A



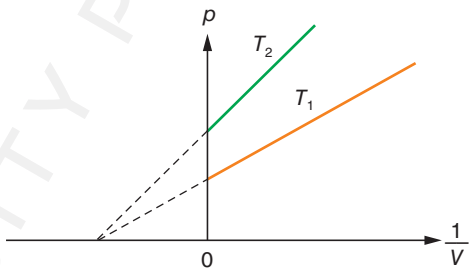
B



C

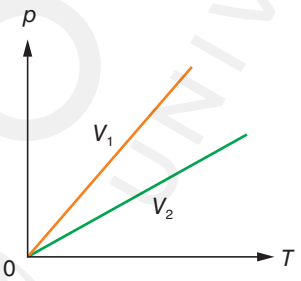


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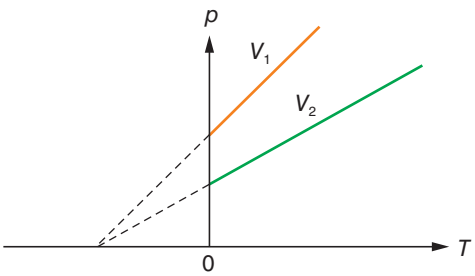


2 Which of the following graphs correctly shows the relationship between the pressure p of a fixed mass of an ideal gas and its temperature T at two different volumes V_1 and V_2? It is given that $V_1 > V_2$.

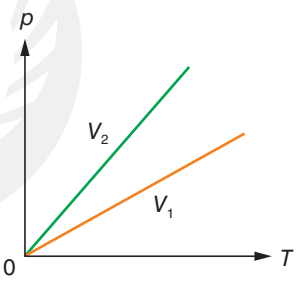
A



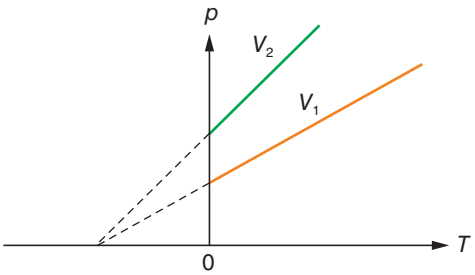
B



C



D



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Ext

☆ HKDSE 2012 Paper 1A Q3

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Slope and intercept

An ideal gas is held inside a container with a movable piston (Fig a). Figure b shows how the volume V of the gas changes with the Celsius temperature T under constant pressure p. The graph meets the temperature axis at Y when extrapolated.

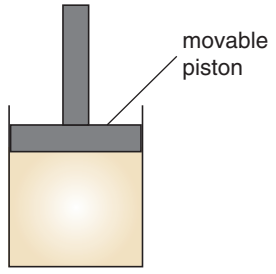


Fig a

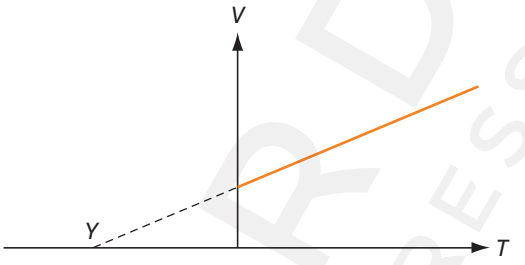


Fig b

Which of the following statements about the graph is/are correct?

(1) It will not meet the temperature axis at Y if the number of ideal gas molecules is reduced.

(2) It will not meet the temperature axis at Y if another ideal gas is used.

(3) Its slope will decrease if p is increased to a higher constant value.

A (1) only

B (3) only

C (1) and (2) only

D (2) and (3) only

Solution

By $pV = nRT$,

we have $V = \left(\frac{nR}{p}\right)T$, where T is the Kelvin temperature of the gas.

Therefore, the V–T graph of any ideal gas should be a straight line meeting the temperature axis at 0 K or –273 °C, depending on the temperature scale used.

∴ (1) and (2) are incorrect.

If p is increased,

the slope of the graph, which is equal to $\frac{nR}{p}$, will decrease.

∴ (3) is correct.

∴ The answer is B.

Common mistake ⚠
Students may wrongly think that the intercept on the temperature axis can be changed. In fact, for an ideal gas, the intercept on the temperature axis is always equal to absolute zero.

▶ Revision 5 Q12 (p.184)

Ext

Exercise 5.1

Take $R = 8.31$ J mol^{-1} K^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1}.

(For Q1–2.) A student studies the pressure law with a gas of a fixed mass and volume. He plots a graph according to the results (Fig a).

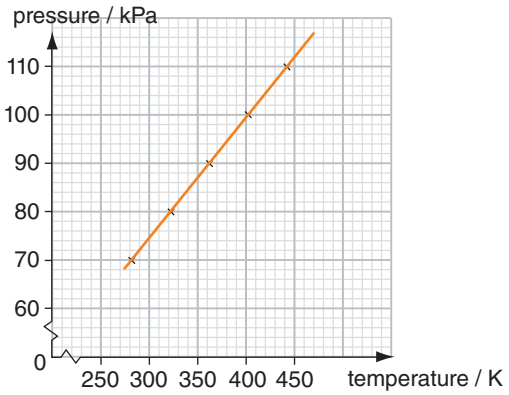


Fig a

1 What is the temperature of the gas if the pressure is 72 kPa?

A 17 °C B 72 K
C 290 °C D 563 °C

2 What is the pressure of the gas if the temperature is 600 K?

A 34.8 kPa B 110 kPa
C 150 kPa D 157 kPa

3 Which of the following statements is/are correct?

(1) Absolute zero is equal to 0 °C.
(2) The difference between the steam point and the ice point is 100 K.
(3) The steam point is equal to 373 K.

A (1) only
B (2) only
C (2) and (3) only
D (1), (2) and (3)

***4** Under which of the following conditions does a real gas behave like an ideal gas?

	Temperature	Pressure
A	high	high
B	high	low
C	low	high
D	low	low

***5** A student studies Boyle's law for an ideal gas and obtains the following results (Fig b).

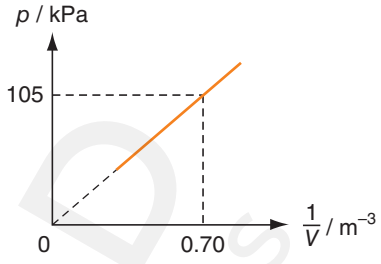


Fig b

The temperature is kept at 20 °C throughout the process. Find the number of moles of the gas.

A 29 mol
B 62 mol
C 295 mol
D 602 mol

***6** Two rigid containers containing the same ideal gas are connected by a short narrow tube as shown (Fig c). The tap is closed initially.

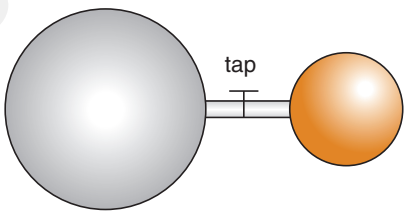


Fig c

Which of the following quantities will become the same after the tap is opened?

(1) The pressure in each container
(2) The number of moles of gas in each container
(3) The volume of gas in each container

A (1) only
B (1) and (2) only
C (1) and (3) only
D (2) and (3) only

7 (a) Calculate the pressure exerted on the ground when a girl with a weight of 500 N stands on the ground with her sports shoes. The total area of her sports shoes in contact with the ground is 10^{-2} m^2.

(b) How would the pressure change if the girl wears high heels with total contact area of 2×10^{-4} m^2?

Ext

8 A marshmallow has a porous structure, i.e. it contains many tiny air holes. Suppose a marshmallow is placed inside a bell jar (Fig d), and then air is pumped out of the jar.



→ This video demonstrates the growing marshmallow.

Video

Fig d

(a) What happens to the marshmallow?

(b) Explain the result by using Boyle's law.

9 A sealed container holds a gas at 300 K. For its pressure to triple, to what temperature should it be heated?

10 A syringe is filled with air of volume 40 cm^3 at 100 kPa and is then sealed. It is compressed so that its volume is reduced by 45%. Find the pressure inside it. Assume that the temperature of the air remains unchanged.

★**11** After conducting an experiment to study Charles' law, a student plots a graph showing how the volume V of an air column varies with temperature T at constant pressure (Fig e).

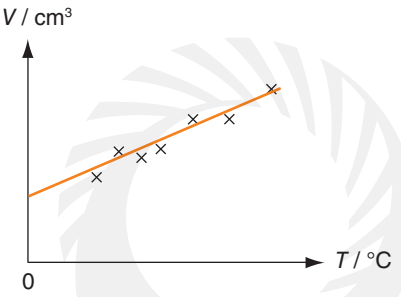


Fig e

(a) Describe how the student can determine the absolute zero from this graph.

(b) The straight line graph passes through points X and Y.

X: T = 0 °C, V = 5.6 cm^3

Y: T = 15 °C, V = 5.9 cm^3

Estimate the absolute zero.

★**12** A coat is put in a storage bag and sealed. The bag contains 1500 cm^3 of air at 100 kPa when sealed. If 70% of the air in the bag is drawn out at a constant temperature (Fig f), what is the new volume of the air inside the bag?



Fig f

★**13** Figure g shows a toy gun and its soft bullet. The body of the gun is hollow and elastic. When the bullet is fixed at the front, it seals the front opening of the gun barrel. Explain why the bullet is expelled when the handle of the gun is squeezed.



→ This video demonstrates how the air toy gun works.

Video

Fig g

★**14** In an experiment that studies Boyle's law, a fixed mass of gas is kept at a constant temperature. Figure h shows how the pressure p of the gas changes with the reciprocal of its volume V.

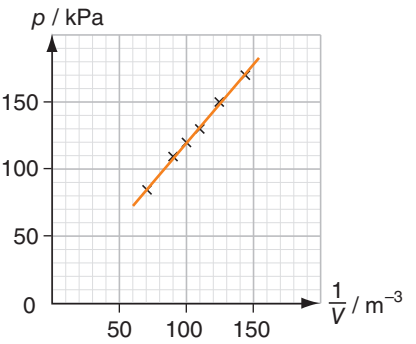


Fig h

(a) What is the volume of the gas if the pressure is 120 kPa?

(b) Verify Boyle's law using the information given in Figure h.

(c) Suppose the experiment is repeated under a lower temperature. Sketch a possible p–$\frac{1}{V}$ graph obtained in Figure h.

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Ext

*15 A student conducts an experiment to study a fixed mass of ideal gas. Figure i shows the p–V graph of the gas at temperatures T_1 and T_2.

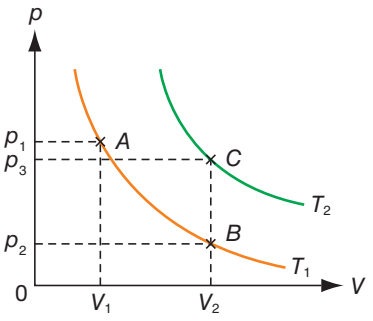


Fig i

(a) The gas is initially at state A, with pressure p_1 and volume V_1. It expands to state B as shown. Express p_2 in terms of p_1, V_1 and V_2. Which gas law do you apply?

(b) The gas is then heated from state B to state C. The final temperature is T_2. Express p_3 in terms of T_1, T_2 and p_2. Which gas law do you apply?

*16 When verifying Charles' law for an ideal gas, a student obtains the following graph (Fig j).

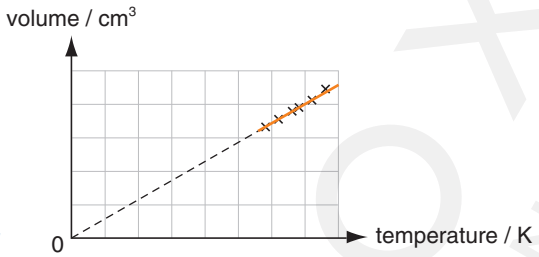


Fig j

Suppose the student repeats the experiment with half of the original amount of gas but with the pressure doubled. Sketch the expected graph of the volume against the temperature for the gas in Figure j. Explain your answer briefly.

*17 Two identical containers, X and Y, are connected by a tube of negligible volume. They are filled with the same type of ideal gas at the same temperature. The flow of gas is blocked by a tap on the tube. X contains 1.5 mol of gas and Y contains 2.4 mol of gas. The pressure in Y is 128 kPa. The tap is then opened so that the gas flows between the containers. The temperature is kept constant throughout the process.

(a) Find the pressure in X before the tap opens.

(b) Find the final pressure in X after the tap is opened.

*18 A syringe contains an ideal gas of 100 cm^3 at 100 kPa.

(a) Given that the ideal gas is at 25 °C, calculate the number of moles and the number of molecules inside the syringe.

(b) The piston of the syringe is pushed slowly to eject gas molecules at a constant temperature and pressure. During this process, the volume of the gas decreases at a constant rate of 10^{-6} m^3 s^{-1}. Find the number of gas molecules ejected per second.

*19 John was challenged to put a shelled hard-boiled egg into a bottle without crushing the egg. The bottle had a long bottleneck and a small opening which looked too small for the egg to pass through (Fig k). Since the egg was deformable and had a smooth surface, John tried to push it directly into the bottle.

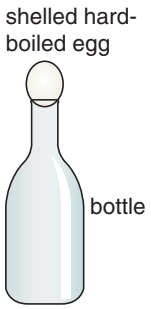


Fig k

(a) The deeper the egg went through the bottleneck, the more difficult it was to push it. Explain the reason.

(b) Mary suggests to change the temperature of the air inside the bottle to make the egg easier to go inside. How should the temperature be changed? Explain your answer.

*20 A student studies the pressure–temperature relationship of an ideal gas. The results are shown in Figure l.

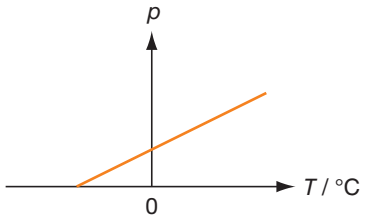


Fig l

(a) The straight line does not pass through the origin. Estimate the intercept on the temperature axis.

(b) Write an expression for the slope.

(c) The student repeats the experiment with a larger amount of the same ideal gas. Sketch the new p–T graph in Figure l.

Ext

5.2 The kinetic theory of gases

1 Explaining gas laws

a Molecular motion and gas pressure

DSE goal

Realize the random motion of molecules in a gas.

Realize the gas pressure resulted from molecular bombardment.

Interpret gas expansion in terms of molecular motion.

Solve problems involving kinetic theory.

DSE exam

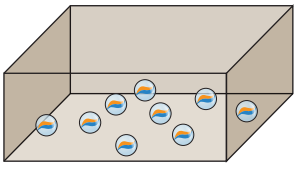
12(1B)Q2(b), 14(1B)Q2(b), 18(1B)Q2(b)(ii), 19(1A)Q3, 21(1B)Q2(b)(ii), 23(1B)Q2(a), (b)

The gas laws only **describe what happens** to a gas under certain conditions but **do not explain why** it behaves in such way. Therefore, scientists proposed the **kinetic theory** of gases to explain the macroscopic properties of gases, such as pressure, volume and temperature, through the microscopic behaviour of the gas molecules.

The kinetic theory is also called the particle theory, which you learned in Secondary 1.

According to the kinetic theory, a gas is made up of a vast number of identical tiny **molecules in random motion all the time** (Fig 5.2a). They collide elastically with each other and the surfaces of the container continually. When the molecules hit a surface of the container and bounce backwards, they exert forces on the surface (Fig 5.2b). This produces pressure on the surface which gives rise to gas pressure.

To get some initial ideas about the kinetic theory, you may imagine a large number of high-speed rubber bouncy balls moving inside a box. If your hand became part of the walls, how would it feel?



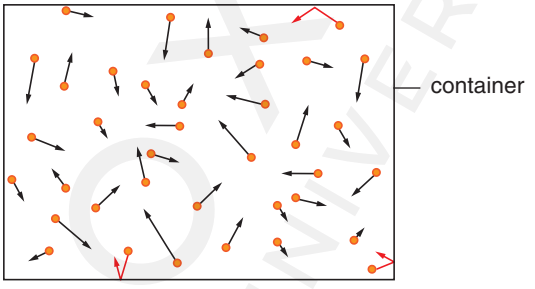


Fig 5.2a *Gas molecules are always in random motion.*

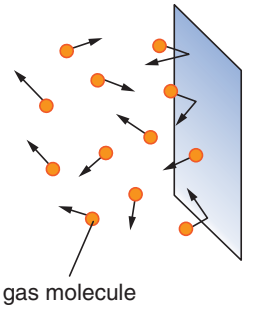


Fig 5.2b *Gas molecules hit a surface and produce gas pressure on it.*

Gas pressure is produced when gas molecules collide with a surface.

Note that if a gas is trapped in a container, the gas pressure is the same everywhere in the container.

The pressure increases if

1 the molecules hit the surface more frequently, or

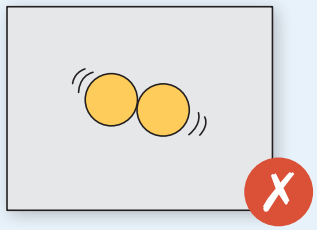
2 the molecules hit the surface more violently, i.e. with a higher speed.

As we have learned in Book 1 Chapter 2, temperature is a measure of the average kinetic energy of the gas molecules. As the temperature rises, the molecules move faster.

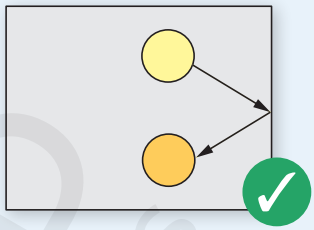
kinetic theory 分子運動論

Ext

Right and wrong



✗ Gas pressure arises due to the collision between gas molecules.



✓ Gas pressure arises due to the collision between gas molecules and a surface.

b Mechanical simulator

We can use kinetic theory to explain the gas laws qualitatively. The discussion can be facilitated with the aid of a mechanical simulator (Fig 5.2c).

A mechanical simulator can simulate the motion of gas molecules. It has a transparent tube containing a large number of ball bearings, which represent gas molecules. The tube is closed by a movable polystyrene piston. A motor-driven vibrator at the bottom of the tube sets the ball bearings in random motion.

The physical quantities in the gas laws can be represented by various features of the simulator as shown in Table 5.2a.

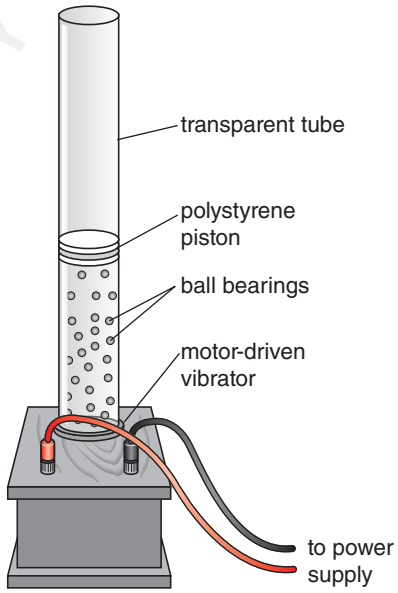


Fig 5.2c *Mechanical simulator of the kinetic theory.*

Physical quantity	Representation in the simulator	Way of adjustment
Pressure	weight of piston	add cardboard disc on top to increase
Temperature	speed of ball bearings	increase the voltage applied to the motor to increase
Volume	height of piston	adjusted automatically until the 'pressure' provided by the ball bearings keeps the piston stable

Table 5.2a *Physical quantities represented by the mechanical simulator.*

Expt

In Experiment 5d, the mechanical simulator is used to facilitate the understanding of the gas laws.

Video

→ This video demonstrates Expt 5d.

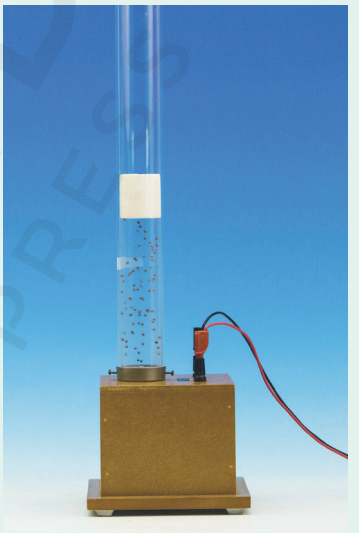
Experiment 5d / Mechanical simulator of the kinetic theory

Set up the mechanical simulator. Study what happens in each of the following cases:

(a) Keep the voltage constant and increase the weight of piston.

(b) Increase the voltage slowly and keep
  (i) the weight of piston constant;
  (ii) the height of piston constant.

(c) Double the number of ball bearings and keep both the voltage and the weight of piston constant.



Discussion

What does each of the above situations demonstrate? See Table 5.2b below.

The experiment illustrates the following situations (Table 5.2b).

Mechanical simulator		Kinetic theory	
Condition	**Result**	**Interpretation**	**Gas law to be simulated**
• voltage unchanged • weight of piston ↑	height of piston ↓	• T is constant. • $p \uparrow \Rightarrow V \downarrow$	Boyle's law (relates p and V)
• height of piston unchanged • voltage ↑	weight of piston ↑	• V is constant. • $T \uparrow \Rightarrow p \uparrow$	pressure law (relates T and p)
• weight of piston unchanged • voltage ↑	height of piston ↑	• p is constant. • $T \uparrow \Rightarrow V \uparrow$	Charles' law (relates V and T)
• voltage and weight of piston unchanged • number of ball bearings n ↑	height of piston ↑	• T and p are constant. • $n \uparrow \Rightarrow V \uparrow$	general gas law (relates V and n)

Table 5.2b *Interpreting the results from the mechanical simulator.*

Note that the mechanical simulator is only a device but not a part of the kinetic theory of gases. It should not be used to replace the qualitative reasoning of the kinetic theory using ideas based on molecular motion.

▶ The mechanical simulator helps us visualize and understand only some quantitative aspects of the kinetic theory. The proper qualitative explanations for the gas laws will be given in the next part.

Ext

Google Forms Microsoft Forms

Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) Gas pressure is produced when gas molecules collide with a surface. (T/F)

(b) Gas pressure is produced when gas molecules collide with each other. (T/F)

2 The voltage in the mechanical simulator of the kinetic theory represents

A the gas pressure.

B the volume of gas.

C the temperature of gas.

D the number of gas molecules.

3 How would Boyle's law be simulated using the mechanical simulator of the kinetic theory? Put a tick in the corresponding box(es).

	Unchanged	Changed
Voltage		
Height of piston		
Weight of piston		
Number of ball bearings		

→ Start Oxford Physics XR 'Kinetic theory of gases' and scan this page to play.

Android iOS

Simulation

→ This simulation shows how the kinetic theory can be used to deduce the gas laws. It shows a dynamic picture of the explanations given in Fig 5.2d to 5.2g. It is available on OUP web in PC version (.exe) only.

C Qualitative explanation using kinetic theory

Using kinetic theory of gases, we can explain the gas laws qualitatively. Let us consider the gas molecules enclosed in a cylinder with a movable piston on its top.

i Boyle's law (pressure–volume relationship)

Since the temperature is kept constant, the average speed of the molecules remains unchanged.

When a gas is compressed, the molecules can only move around in a smaller volume. They hit the cylinder walls more frequently and thus produce a higher pressure (Fig 5.2d).

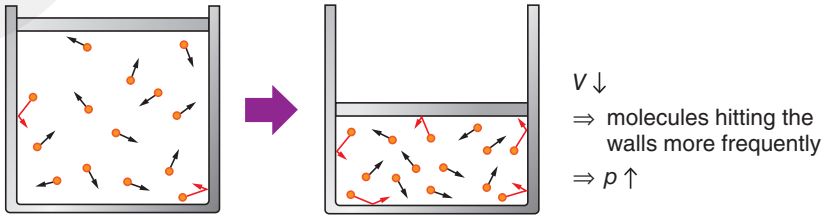


Fig 5.2d *Explaining Boyle's law.*

Ext

ii Pressure law (pressure–temperature relationship)

When temperature rises, the molecules move faster. With the volume fixed, the molecules hit the cylinder walls more frequently and move violently. Therefore, the pressure increases (Fig 5.2e).

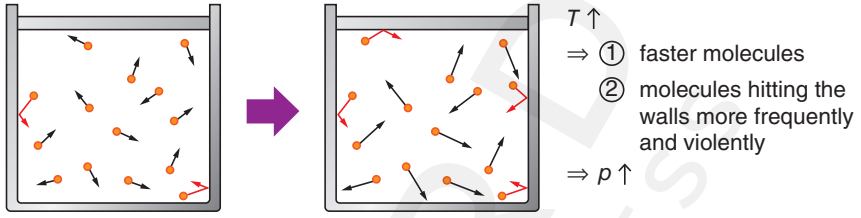


Fig 5.2e *Explaining pressure law.*

iii Charles' law (volume–temperature relationship)

When temperature rises, the molecules move faster. As a result, the molecules hit the cylinder walls more frequently and more violently. If the pressure is to remain constant, the volume must increase so that the frequency of collisions with the walls decreases (Fig 5.2f).

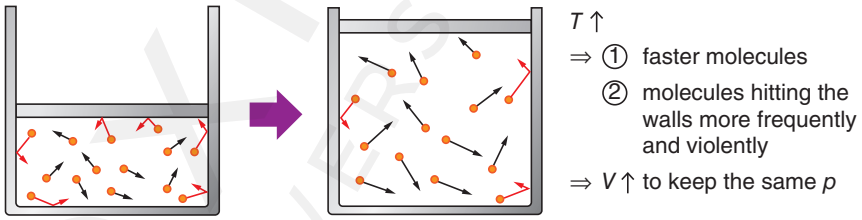


Fig 5.2f *Explaining Charles' law.*

iv Relationship between volume and number of molecules

When the number of molecules increases, the collisions on the wall become more frequent. If the pressure is to remain constant, the volume must increase so that the frequency of collisions with the walls decreases (Fig 5.2g).

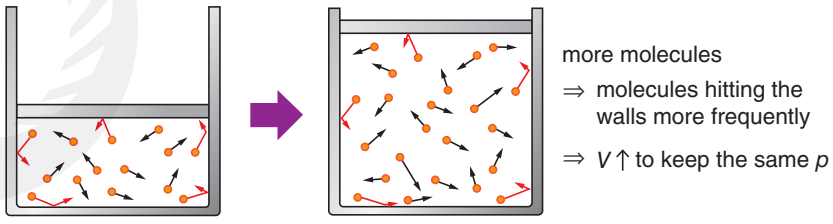


Fig 5.2g *Explaining the relationship between volume and number of molecules.*

Ext

Dangerous! Do not try this!

Writing practice

Question

Dry ice changes directly from solid to gas at a temperature above −78.5 °C. Some dry ice is put into an air-tight rigid bottle at room temperature. Using kinetic theory, explain the increase in gas pressure inside the bottle. (4 marks)

Answer

When the dry ice in the bottle changes to gas, the number of gas molecules in the bottle increases. 1A

The speeds of the molecules also increase due to the increase in temperature. 1A

Since the volume remains unchanged, 1A

the collisions of molecules on the wall of the bottle become more frequent. 1A

Therefore, the gas pressure inside the bottle increases.

Practise yourself

1 The piston of a sealed syringe is pulled outwards slowly (Fig a) so that the temperature change of the gas inside is negligible. How will the gas pressure inside the syringe change? Explain using kinetic theory. (4 marks)

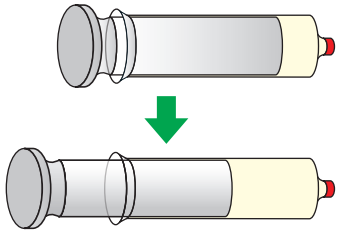


Fig a

2 How will the gas pressure inside a sealed syringe change if it is put in a beaker of cold water with the position of the piston fixed (Fig b)? Explain using kinetic theory. (4 marks)



Fig b

Ext

DSE goal

State the assumptions of the kinetic model of an ideal gas.

2 Assumptions about ideal gas

The kinetic theory offers explanations on the behaviour of an ideal gas as predicted by the gas laws. The theory is based on the following assumptions about the molecules of an ideal gas.

1 **All the molecules are identical and have the same mass.**
2 **All the molecules are in continual random motion.**
3 **The number of molecules in the container is large.**
4 **The size of each molecule is negligible compared with the separation between them.**
5 **The duration of a collision is negligible compared with the time between collisions.**
6 **The collisions between molecules and the container and between the molecules are perfectly elastic.**
7 **Intermolecular forces are negligible.**

✎ In a real gas, the molecules may not be identical. Also, there are intermolecular forces between the molecules.

A real gas does not satisfy all the assumptions above and therefore does not follow gas laws precisely. However, as mentioned on p.165, at **low pressures and high temperatures**, the approximation is good enough and a real gas behaves like an ideal gas.

From a macroscopic point of view, a gas that satisfies the general gas law is an ideal gas. Microscopically, a gas is ideal if the gas molecules satisfy all of the above assumptions.

DSE goal

Realize $pV = \frac{Nm\overline{c^2}}{3}$ that connects microscopic and macroscopic quantities of an ideal gas and solve problems.

Interpret temperature of an ideal gas using $KE_{average} = \frac{3RT}{2N_A}$.

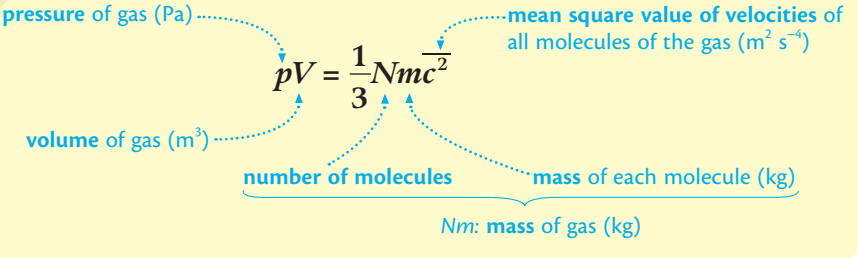
DSE exam

13(1A)Q3, 16(1A)Q3, 17(1B)Q3, 21(1A)Q3, 22(1B)Q2

3 Relations and expressions derived from the kinetic theory of gases

a Pressure and molecular motion of a gas

Consider a container of volume V holding N gas molecules and each molecule has a mass m. The molecules move at different velocities c. Based on the seven assumptions above, the gas pressure p can be derived. The following relationship between the various quantities will be obtained:



From this equation we can find $\sqrt[3]{\frac{V}{N}}$, which is the average separation of gas molecules. ▶

✎ Refer to the supplementary material on OUP web for the derivation of this equation. (Select 'Lesson Preparation' then click 'Supplementary Materials')

Ext

However, the general gas law $pV = nRT$ contains the macroscopic properties of a gas on both sides.

▶ The left-hand side of the above equation contains the macroscopic properties of a gas whereas the right-hand side contains the microscopic parameters from the kinetic theory.

The term $\overline{c^2}$ is called the *mean square value* of the velocities of all gas molecules. It is calculated by taking the average of the squared values of the velocities of all the gas molecules in the container, i.e.

$$\overline{c^2} = \frac{c_1^2 + c_2^2 + \ldots + c_N^2}{N}$$

Molecules move at different velocities. Instead of considering an individual molecule, we take the average value of all molecules.

▶ where c_1, c_2, …, c_N are the velocities of the N gas molecules.

The product Nm is the mass of the gas.

Since Nm is the total mass of the gas and V is the volume, the expression $pV = \frac{1}{3}Nm\overline{c^2}$ may also be written as

$$p = \frac{1}{3}\rho\overline{c^2}$$

where ρ is the density of the gas.

Know more

Conversion between different quantities (I)

There are different ways to describe the mass or the number of molecules of a gas. They are denoted by similar symbols.

Quantity	Physical meaning
n	Number of moles of gas molecules
N	Number of gas molecules
N_A	Number of molecules in one mole of gas = 6.02×10^{23} mol^{-1}
m	Mass of each gas molecule
Nm	Mass of all molecules in a gas = mass of the gas
N_Am	Mass of one mole of gas molecules = molar mass of the gas

Therefore, $n = \frac{N}{N_A}$.

b Temperature and kinetic energy of gas molecules

Combine the equation $pV = \frac{1}{3}Nm\overline{c^2}$ and the general gas law $pV = nRT$, we get:

$$\frac{1}{3}Nm\overline{c^2} = nRT$$

$$\left(\frac{2}{3}\right)\frac{1}{2}Nm\overline{c^2} = nRT$$

$$\frac{1}{2}Nm\overline{c^2} = \frac{3}{2}nRT$$

By $\overline{c^2} = \frac{c_1^2 + \ldots + c_N^2}{N}$,

$\frac{1}{2}Nm\overline{c^2}$

$= \frac{1}{2}mc_1^2 + \ldots + \frac{1}{2}mc_N^2$

= total kinetic energy

▶ The left-hand side of the equation, $\frac{1}{2}Nm\overline{c^2}$, is simply the total kinetic energy of all the gas molecules. Therefore,

$$\text{total kinetic energy of a gas} = \frac{3}{2}nRT$$

mean square value 方均值

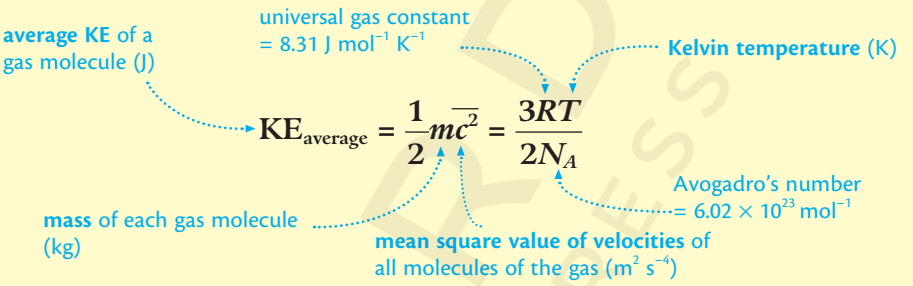
Ext

See Know more on p.175. ▶

$$\text{total kinetic energy of one mole of gas molecules} = \frac{\frac{3}{2}nRT}{n} = \frac{3}{2}RT$$

Recall that $N = nN_A$. ▶

$$\text{average kinetic energy of a gas molecule} = \frac{\frac{3}{2}nRT}{N} = \frac{3RT}{2N_A}$$



This means the Kelvin temperature of a gas is directly proportional to the average molecular kinetic energy, i.e.

$$T \text{ (in kelvin)} \propto KE_{average}$$

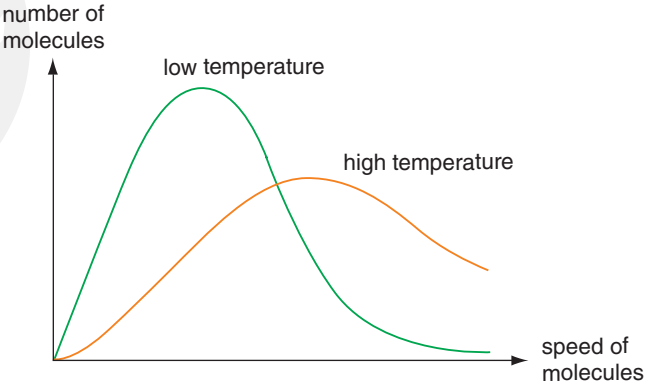
Temperature is a measure of the average kinetic energy of the gas molecules.

According to assumption 7 on p.172, since the gas molecules do not interact with each other, they do not have molecular PE. Hence, the total molecular KE is also equal to the internal energy of the gas.

$$\therefore \text{ Internal energy of a gas} = \frac{3}{2}nRT$$

C Root-mean-square speed of gas molecules

The molecules in a gas move at a wide range of speeds (Fig 5.2h). The distribution in speeds depends on temperature.



The distribution shows how many molecules (vertical axis) are moving at each particular speed (horizontal axis) under a certain temperature. If the curve is higher at certain speeds, it means more molecules are moving at those speeds. ▶

Fig 5.2h *Distribution of molecular speeds.*

Ext

Consider the formula $pV = \frac{1}{3}Nm\overline{c^2}$ and $\frac{1}{2}m\overline{c^2} = \frac{3RT}{2N_A}$. In these equations, $\overline{c^2}$ is the mean value of c^2 of all the gas molecules. The typical speed of molecules can be found by taking the square root of $\overline{c^2}$. This is called the **root-mean-square speed**, denoted by c_{rms}.

It is sometimes written as r.m.s. speed for short. ▶

By $pV = \frac{1}{3}Nm\overline{c^2}$,

$$c_{rms} = \sqrt{\overline{c^2}} = \sqrt{\frac{3pV}{Nm}}$$

By $\frac{1}{2}m\overline{c^2} = \frac{3RT}{2N_A}$,

$$c_{rms} = \sqrt{\overline{c^2}} = \sqrt{\frac{3RT}{mN_A}}$$

Note that Nm is the total mass of the gas, and mN_A is the mass of 1 mole of the gas, i.e. the **molar mass** of the gas.

Also, different gases have different molar masses and hence different c_{rms}. For example, at room temperature, the c_{rms} of the gas molecules of hydrogen (H_2) and oxygen (O_2) are 1920 m s^{-1} and 482 m s^{-1} respectively.

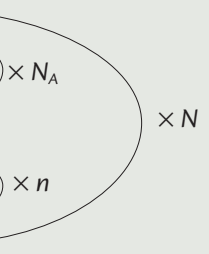
Know more

Conversion between different quantities (II)

The table below summarizes different conversions.

Note that the unit of molar mass is g mol^{-1} or kg mol^{-1} and the unit of the molecular kinetic energy of one mole of gas is J mol^{-1}.

	Mass	KE
1 molecule	m	$\frac{3}{2}\frac{1}{N_A}(RT)$
1 mole	N_Am	$\frac{3}{2}(RT)$
Total (n mole)	nN_Am (or Nm)	$\frac{3}{2}n(RT)$



root-mean-square speed 方均根速率　　molar mass 摩爾質量

Ext

Example 9 | Root-mean-square speed

An ideal gas is sealed in a container of fixed volume. The gas is heated so that the root-mean-square speed c_{rms} of the gas molecules is doubled. The gas changes its temperature from T_1 to T_2 and its pressure from p_1 to p_2.

(a) Find p_2 in terms of p_1. It is given that $c_{rms} = \sqrt{\frac{3pV}{Nm}}$.

(b) Find T_2 in terms of T_1. It is given that $c_{rms} = \sqrt{\frac{3RT}{mN_A}}$.

Solution

(a) By $c_{rms} = \sqrt{\frac{3pV}{Nm}}$, for constant V, N and m,

$$\frac{\sqrt{p_2}}{\sqrt{p_1}} = \frac{2c_{rms}}{c_{rms}} = 2$$

$$p_2 = 4p_1$$

(b) By $c_{rms} = \sqrt{\frac{3RT}{mN_A}}$, for constant m,

$$\frac{\sqrt{T_2}}{\sqrt{T_1}} = \frac{2c_{rms}}{c_{rms}} = 2$$

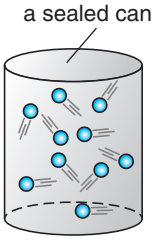
$$T_2 = 4T_1$$

Think about...

If the container is allowed to expand during the heating of the gas, will the answers to (a) and (b) change?

Example 10 | Gas in a can

A sealed can of volume 350 cm^3 contains an ideal gas at 115 kPa and 20 °C. Take the molar mass of the gas to be 29.0 g mol^{-1}, and R to be 8.31 J mol^{-1} K^{-1}.



(a) (i) Find the number of moles of gas in the can.

(ii) Find the root-mean-square speed of the gas molecules.

(iii) Find the kinetic energy of one mole of the gas molecules.

(b) The can is now placed in cold water and the gas inside is cooled to 5 °C. Find the change in the total kinetic energy of the gas molecules.

Ext

Tip
Write down the data provided in the question to determine the equation that should be applied.

1 cm^3 = 1 × 10^{-6} m^3

Tip
Express mN_A in kg mol^{-1}.

Solution

(a) $V = 350\ \text{cm}^3$, $p = 115\ \text{kPa}$, $T = 20\ ^\circ\text{C}$, $mN_A = 29.0\ \text{g mol}^{-1}$

(i) By $pV = nRT$,

$$n = \frac{pV}{RT} = \frac{(115 \times 10^3)(350 \times 10^{-6})}{8.31 \times (20 + 273)} = 0.0165\ \text{mol}$$

The number of moles of the gas is 0.0165 mol.

(ii) By $\frac{1}{2}m\overline{c^2} = \frac{3RT}{2N_A}$,

$$\sqrt{\overline{c^2}} = \sqrt{\frac{3RT}{mN_A}} = \sqrt{\frac{3 \times 8.31 \times (20 + 273)}{29.0 \times 10^{-3}}} = 502\ \text{m s}^{-1}$$

The root-mean-square speed of the gas molecules is 502 m s^{-1}.

(iii) KE of one mole of the gas

$$= N_A \times \frac{3RT}{2N_A} = \frac{3}{2}RT = \frac{3}{2} \times 8.31 \times (20 + 273) = 3650\ \text{J mol}^{-1}$$

(b) Total KE of the gas molecules

$$= N \times \frac{1}{2}m\overline{c^2} = N \times \frac{3RT}{2N_A} = nN_A \times \frac{3RT}{2N_A} = \frac{3}{2}nRT$$

Change in total KE

$$= \frac{3}{2}nR(T_2 - T_1) = \frac{3}{2} \times 0.0165 \times 8.31 \times (5 - 20) = -3.09\ \text{J}$$

Google Forms Microsoft Forms

Quick check 7

→ E-version in OUP Exercise Platform is also available.

1 Which of the following quantities must increase when the temperature of an ideal gas increases?

- ☐ The pressure of the gas
- ☐ The root-mean-square speed of the gas molecules
- ☐ The average kinetic energy of the gas molecules
- ☐ The internal energy of the gas

2 If m is the mass of the molecule of a gas and n is the number of moles of the gas molecules, which of the following is the molar mass of the gas?

A mn B mN_A C nN_A

3 Consider a fixed mass of an ideal gas. Determine how the root-mean-square speed of the gas molecules will change in each of the following cases.

(a) The gas is heated under constant volume.

(b) The gas expands under constant pressure.

(c) The gas is compressed under constant temperature.

Ext

Learn by practice 5

Take $R = 8.31$ J mol^{-1} K^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1}.

1 The pressure of an ideal gas inside a container of 40 cm^3 is 120 kPa. The total mass of the gas is 6.5×10^{-5} kg. Find the mean square speed of the gas molecules.

2 Show that the root-mean-square speeds of H_2 and O_2 gas molecules at room temperature 25 °C are 1920 m s^{-1} and 482 m s^{-1} respectively. Given: molar mass of H_2 = 2.016 g mol^{-1}, molar mass of O_2 = 31.999 g mol^{-1}

3 A box contains 0.5 mole of an ideal gas at 25 °C. Find

(a) the total kinetic energy of all the gas molecules,

(b) the kinetic energy of one mole of the gas,

(c) the average kinetic energy of each gas molecule.

4 There are 7.28×10^{24} molecules of an ideal gas in a container. The gas is heated from 25 °C to 80 °C. Find the increase in the total kinetic energy of the gas molecules.

Nature of science

Scientific theory

A theory is a 'thought-model' created by scientists to explain and predict certain phenomena. Scientists accept the kinetic theory not because they can actually see the gas molecules, but rather because the theory can explain and predict the behaviours of gases to a satisfactory degree.

In fact, there was another theory that once competed with the kinetic theory:



Bernoulli (18th century)

(Kinetic theory)

A hot object is hot because of its molecular kinetic energy.

(*Caloric theory*)

A hot object is hot because it contains a substance called caloric.



Lavoisier (18th century)



Clausius (19th century)

Hmmm…both theories agree well with observations. There is no reason to reject any of them!

Scientists look for evidence that either supports or rejects a theory. They retain the theory if it agrees with observations. Otherwise, they modify or even discard it.



Maxwell Boltzmann

(19th century)

I've found evidence that agrees with kinetic theory but not the caloric theory.

This means we should reject caloric theory. Let us modify kinetic theory to explain even more observations.



Joule (19th century)

Scientific investigation is a continuous process in which scientists revise or discard existing theories to come up with better theories with greater explanatory and predicting power. The perseverance and diligence of scientists have enabled us to develop a profound understanding of nature.

Ext

☆ HKDSE 2016 Paper 1B Q2

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Pumping a bicycle tyre

Air is pumped into a bicycle tyre with a pump via a rubber tubing. In each stroke, 150 cm^3 of air at an atmospheric pressure of 100 kPa enters the tyre. Assume the air temperature is kept constant at 20 °C throughout the pumping process.



(a) The tyre is pumped to its maximum volume of 1200 cm^3 with the air inside kept at atmospheric pressure. Estimate the number of air molecules trapped inside the tyre. (3 marks)

(b) Using the kinetic theory, explain why the pressure inside the tyre increases when air is kept being pumped into the tyre after the tyre's volume has reached 1200 cm^3. (3 marks)

(c) After the tyre's volume has reached 1200 cm^3, three more strokes of the pump are completed. Find the final pressure (in kPa) inside the tyre. (3 marks)

Solution

(a) By $pV = nRT$,

$$n = \frac{pV}{RT} = \frac{(100 \times 10^3)(1200 \times 10^{-6})}{(8.31)(20 + 273)} \quad \text{1M}$$

$$= 0.0493 \text{ mol}$$

Number of air molecules $= nN_A = 0.0493 \times (6.02 \times 10^{23})$ 1M

$= 2.97 \times 10^{22}$ 1A

Common mistake ⚠
Students may wrongly think that the physical quantity required is *n*, which is the no. of moles of air.

(b) When air is kept being pumped into the tyre, the number of air molecules inside the tyre increases. 1A

Since the volume and the temperature remain unchanged, 1A

the collisions of molecules on the wall of the tyre become more frequent. 1A

Common mistake ⚠
Students may fail to state this condition.

Therefore, the pressure inside the tyre increases.

(c) Number of moles of air molecules pumped in each stroke

$$= \frac{pV}{RT} = \frac{(100 \times 10^3)(150 \times 10^{-6})}{(8.31)(20 + 273)} = 0.006\,16 \text{ mol} \quad \text{1M}$$

Number of moles of air molecules inside the tyre after three more strokes are completed

$= 0.0493 + 3 \times 0.00616 = 0.0678$ mol

By $pV = nRT$,

$$p = \frac{nRT}{V} = \frac{0.0678 \times 8.31 \times (20 + 273)}{1200 \times 10^{-6}} \quad \text{1M}$$

$= 138\,000$ Pa

$= 138$ kPa 1A

▶ Revision 5 Q29 (p.190)

Ext

Exercise 5.2

Take $R = 8.31$ J mol^{-1} K^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1}.

1 Which of the following is **not** an assumption made about an ideal gas in the kinetic theory?

A Molecules are in random motion.

B All collisions are perfectly elastic.

C The speeds of all molecules are the same at a given temperature.

D Intermolecular forces are negligible.

2 A sealed container of volume 100 cm^3 contains an ideal gas of pressure 80 kPa. What is the total kinetic energy of all the gas molecules?

A 4 J B 5.33 J

C 8 J D 12 J

★**3** When a bag of potato chips is opened, the pressure inside the bag decreases. Which of the following is/are the correct explanation for this?

(1) The air molecules inside the bag collide with the inner wall of the bag less frequently.

(2) The air molecules inside the bag collide with the inner wall of the bag less violently.

(3) The collisions between the air molecules inside the bag and the inner wall of the bag become inelastic.

A (1) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★**4** The figure below shows the variation of the total molecular KE with the temperature for ideal gases X and Y (Fig a), each in a separate container. Which of the following is the reason why the graph of X has a larger slope?

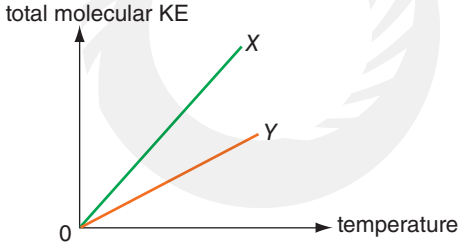


Fig a

A The volume of X is larger.

B The number of molecules in X is larger.

C The pressure of X is larger.

D The mass of the molecules in X is larger.

5 The air inside a sealed tin can is heated.

(a) State what happens to the average speed of the air molecules.

(b) Explain the change in the air pressure inside the can using the kinetic theory.

★**6** (a) Given that the molar mass of an ideal gas is 0.0337 kg mol^{-1}, find the mean square speed of the gas molecules at 25 °C.

(b) The gas is then heated up and expands with constant pressure. The number of molecules is unchanged. How does each of the following quantities change?

(i) Mean square speed of the gas molecules

(ii) Volume of the gas

★**7** Consider an ideal gas with a fixed mass. In each of the following cases, what happen to the average kinetic energy of the gas molecules?

(a) The volume of the gas triples at constant pressure.

(b) The volume of the gas triples at constant temperature.

★**8** From the kinetic theory of gases, we obtain:

$$pV = \frac{1}{3}Nmc^{\overline{2}}$$

(a) State the physical meaning of m and Nm.

(b) A fixed mass of an ideal gas is compressed to half of its original volume at constant pressure. Estimate the ratio of the root-mean-square speed of the gas molecules before the compression to that after the compression.

★**9** Both the air temperature and the atmospheric pressure decrease when altitude increases. When a helium balloon is let go and rises, it expands. Assume that the helium inside the balloon is always at the same temperature as the outside.

(a) How does the average kinetic energy of the helium molecules change when the balloon rises?

(b) Using the kinetic theory, explain how the pressure inside the balloon changes when the balloon rises.

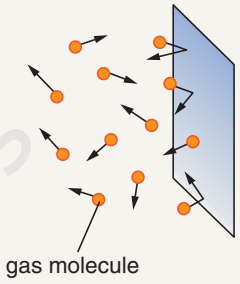
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Ext

Key concepts 5

1 Pressure *p*

- $p = \frac{F}{A}$
- A force gives rise to a greater pressure on a smaller surface.
- Unit: Pa (1 Pa = 1 N m^{-2})
- Gas pressure: produced when gas molecules collide with a surface



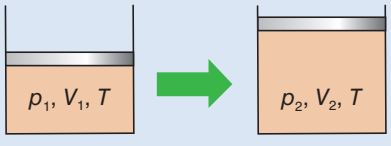
Q28 Exam Report: This question tested candidates' knowledge and understanding on gas laws and kinetic theory. The general performance was fair.

(a) Many candidates did not fully understand the relationship between pressure and volume in the context of pumping a ball. Some were able to relate the pressure with the number of gas molecules in the basketball, but failed to work out their proportional relationship with the required volume of air originally at atmospheric pressure.

(b) Few candidates were able to state that both temperature and volume were constant before they went on to explain the increase of pressure inside the basketball using kinetic theory.

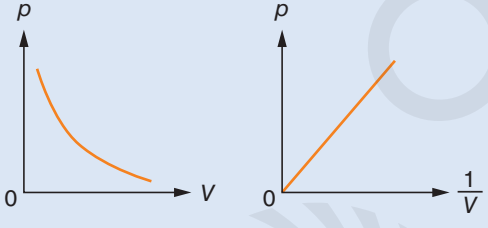
2 Gas laws

Boyle's law

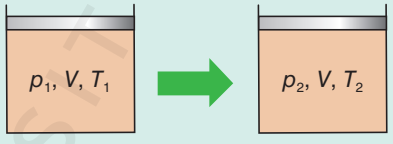


For a fixed mass of gas with constant T,

$$p_1V_1 = p_2V_2$$



Pressure law (*p*–*T* relationship)

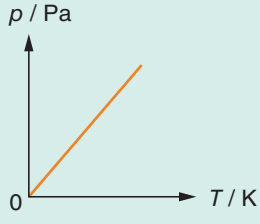


- For a fixed mass of gas with constant V,

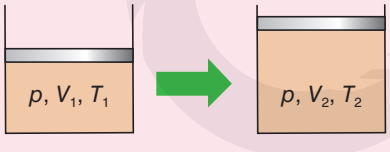
$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

where T_1 and T_2 are measured in kelvins (K).

- Kelvin temperature = Celsius temperature + 273
- Absolute zero
  = 0 K
  = −273 °C



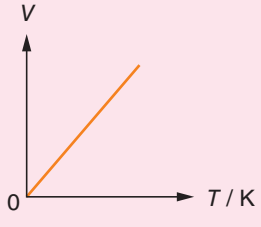
Charles' law (*V*–*T* relationship)



For a fixed mass of gas with constant p,

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

where T_1 and T_2 are measured in kelvins (K).



Q29 Exam Report: This question tested candidates' knowledge and understanding on gas laws. The general performance was fair.

(a) Most candidates were able to find the number of moles though some of them failed to get the final answer which is the number of gas molecules.

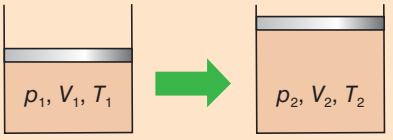
(b) (i) Some had a misconception that the gas temperature could be kept constant by wrapping the syringe with an insulator.

(ii) Quite a number of them wrongly thought that V_0 is the volume of gas at absolute zero or under infinite pressure.

(iii) This part was well answered.

Ext

General gas law



For a fixed mass of gas, $\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$

Besides, for n moles of gas with pressure p, volume V and Kelvin temperature T,

$pV = nRT$ (where R = 8.31 J mol^{-1} K^{-1})

Ideal gas

- A gas that obeys the gas laws
- A real gas behaves like an ideal gas at
  - low pressure
  - high temperature

Q30 Exam Report: The question was in general well answered.

(a) Most managed to obtain the answer despite some forgetting to convert the temperature given in Celsius to an absolute temperature on the Kelvin scale or mixed up number of molecules with the number of moles.

(b) (i) Many candidates knew that the decrease in pressure would in effect increase the volume of the balloon. However, few were able to compare the change in pressure with the change in temperature while some wrongly thought that the volume remained unchanged.

3 Kinetic theory of gases

Assumptions about ideal gas

1 All the molecules are identical and have the same mass.
2 All the molecules are in continual random motion.
3 The number of molecules in the container is large.
4 The size of each molecule is negligible compared with the separation between them.
5 The duration of a collision is negligible compared with the time between collisions.
6 The collisions between molecules and the container and between the molecules are perfectly elastic.
7 Intermolecular forces are negligible.

(ii) Candidates did well in the calculation although some considered the temperature to be constant and wrongly used Boyle's law to estimate the pressure. A few made mistakes when converting units or wrongly took the base of the exponential function as an electronic charge.

Relating macroscopic and microscopic properties

- $pV = \frac{1}{3}Nm\overline{c^2}$
- $\text{KE}_{\text{average}} = \frac{3RT}{2N_A} \propto T$

Key terms

1	absolute temperature scale 絕對溫標	p.147
2	atmospheric pressure 大氣壓強	p.140
3	Avogadro's number 阿佛加德羅數	p.155
4	Bourdon gauge 布爾登氣壓計	p.142
5	Boyle's law 波義耳定律	p.144
6	Charles' law 查理定律	p.152
7	general gas law 普適氣體定律	p.155
8	ideal gas 理想氣體	p.156
9	kelvin (K) 開	p.147
10	Kelvin temperature scale 開氏溫標	p.147
11	kinetic theory 分子運動論	p.166
12	molar mass 摩爾質量	p.175
13	mole (mol) 摩爾	p.155
14	pascal (Pa) 帕斯卡	p.138
15	pressure 壓強	p.138
16	pressure law 氣壓定律	p.149
17	root-mean-square speed 方均根速率	p.175
18	universal gas constant 普適氣體常數	p.155

Ext

Revision 5

Take $R = 8.31$ J mol^{-1} K^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1}.

Multiple-choice questions

★1 5.1 Marcus finds it easier to walk on snow with snowshoes (Fig a). Without snowshoes, his feet sink into the snow (Fig b).



Fig a



Fig b

Which of the following best explains why snowshoes can help him walk on snow?

A The area of contact between the snowshoes and the snow is larger than that of ordinary shoes.

B Snowshoes exert a smaller force on the snow.

C Snowshoes exert a higher pressure on the snow.

D Snowshoes have a smoother contact surface with the snow.

★2 5.2 A gas satisfies the equation $pV = \frac{1}{3}Nm\overline{c^2}$. Which of the following statements is/are correct?

(1) The gas has $\frac{N}{N_A}$ moles of molecules.

(2) Every molecule moves at the same speed c.

(3) The mass of the gas is Nm.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★3 5.1 Two containers X and Y are filled with two different ideal gases. Under atmospheric pressure, the gas in X has a volume of 1.2 m^3 at 20 °C and the gas in Y has a volume of 0.9 m^3 at 25 °C. Calculate the ratio of the number of molecules in X (N_X) to that in Y (N_Y).

A 1.07 : 1 B 1.31 : 1

C 1.36 : 1 D 1.67 : 1

★4 5.1 A helium balloon deflates after a few days (Fig c).



Fig c

Which of the following statements about the deflated balloon is/are correct?

(1) The pressure of the helium gas is smaller than the air pressure outside the balloon.

(2) The helium gas does not fill up all the space inside the balloon.

(3) The volume of the balloon will increase if the air pressure outside the balloon decreases.

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

★5 5.2 At 20 °C, the r.m.s. speed of the molecules in a fixed mass of ideal gas is u. If the temperature is increased to 80 °C, What is the r.m.s. speed of the gas molecules?

A $4u$

B $2u$

C $1.20u$

D $1.10u$

★6 5.2 Two identical containers (Fig d) hold two ideal gases X and Y at the same temperature. Which of the following quantities must be the same for the two gases?



Fig d

A The total molecular kinetic energy

B The r.m.s. speed of the molecules

C The average molecular kinetic energy

D The total mass of the gas

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Ext

*7 5.1 Which of the following statements is/are **incorrect**?

(1) No object can be cooled to below absolute zero.

(2) If a body is cooled to absolute zero, the average kinetic energy of its molecules drops to the minimum value.

(3) Absolute zero is 273 °C.

A (3) only
B (1) and (2) only
C (2) and (3) only
D None of the above

**8 5.1 Figure e shows the relationship between the pressure and the temperature of a fixed mass of an ideal gas inside a plastic bag. The gas changes from state X to state Y along path XY. How does the volume of the gas change?

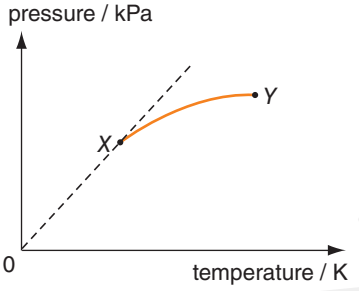


Fig e

A Increases
B Decreases
C Remains unchanged
D Cannot be determined

Refer p.160

(For Q9–10.) The pressure of an ideal gas at state M is increased at a constant volume and then expands at a constant pressure to state N (Fig f).
Given: 1 atm = 10^5 Pa

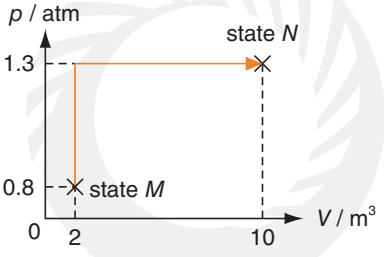


Fig f

**9 5.2 Find the change in the internal energy of the gas.

A 2.40×10^5 J
B 1.71×10^6 J
C 1.95×10^6 J
D 2.19×10^6 J

Refer p.176–177

**10 5.2 The temperature of the gas at state N is 60 °C. Find the number of molecules of the gas.

A 2.61×10^{20}
B 1.57×10^{22}
C 4.70×10^{23}
D 2.83×10^{26}

Refer p.157

**11 5.2 Figure g shows two containers made of conducting walls. One of the containers is filled with ideal gas X and the other is filled with ideal gas Y. Which of the following statements is/are correct after a long period of time?

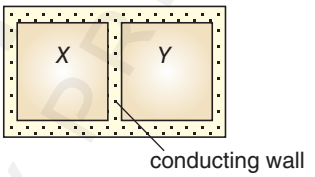


Fig g

(1) The root-mean-square speeds of the molecules of the two gases will become the same even if their molar masses are not the same.

(2) The total kinetic energies of the two gases will become the same even if their masses are not the same.

(3) The average kinetic energies of the molecules of the two gases will become the same even if their number of moles are not the same.

A (1) only
B (2) only
C (3) only
D (2) and (3) only

Refer p.174–175

5.1 **12 HKDSE 2012 Paper 1A Q3**

An ideal gas is contained in a closed vessel of fixed volume. Figure h shows the variation of pressure p of the gas against its Celsius temperature θ.

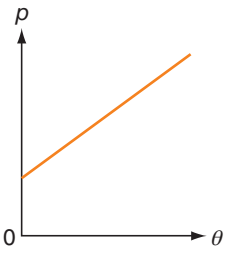
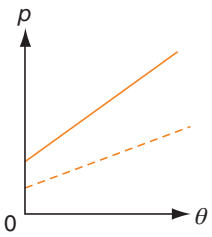


Fig h

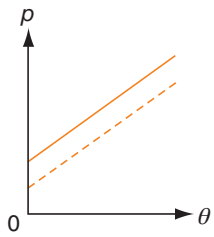
Ext

If the number of gas molecules in the vessel is halved, which graph of the dotted line best shows the relationship between p and θ?

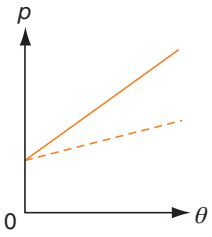
A (38%)



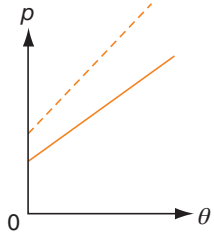
B



C



D



5.1 13 HKDSE 2013 Paper 1A Q4

Vessel X of volume V and vessel Y of volume $2V$ are connected by a short narrow tube as shown (Fig i). Initially, tap S is closed and the same kind of ideal gas at the same temperature is contained in X and Y at pressure $2p$ and p respectively. The tap S is then opened and equilibrium state is finally reached with the temperature unchanged. Which statement is **incorrect**?

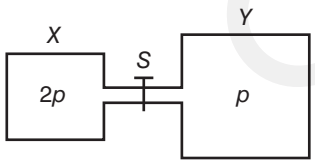


Fig i

A Before S is opened, both vessels contain the same number of gas molecules.

B Before S is opened, the average kinetic energy of the gas molecules in both vessels is the same.

C When S is opened, a net flow of gas from X to Y occurs.

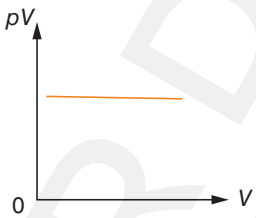
D When equilibrium is reached, the gas pressure becomes $\frac{3}{2}p$. (48%)

Q14 Exam Report: Over 40% of the candidates chose option C to represent proportional relationship, which is incorrect.

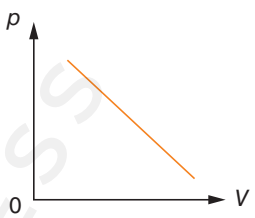
5.1 14 HKDSE 2018 Paper 1A Q4

From which graph below can one deduce that the pressure p of a fixed mass of an ideal gas is inversely proportional to its volume V when the temperature of the gas is kept constant?

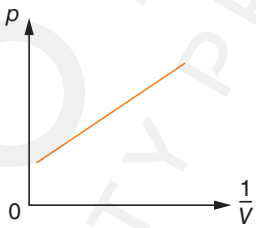
A (11%)



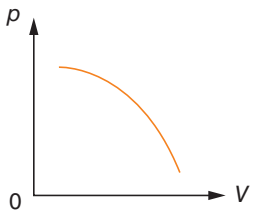
B (35%)



C (43%)



D (11%)



5.1 15 HKDSE 2019 Paper 1A Q3

The figure shows an inexpansible container with an opening (Fig j).

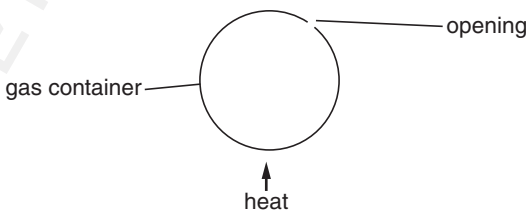


Fig j

When the gas inside the container is being heated slowly by a heater, which statements about the gas molecules in the container are correct?

(1) The number of the molecules decreases.

(2) The average kinetic energy of molecules increases.

(3) The average separation between molecules remains unchanged.

A (1) and (2) only (67%)

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

Ext

Conventional questions

*16 5.1 Meatballs usually contain many air holes. Figure k shows some meatballs in a hotpot.



Fig k

Suppose a meatball at 0 °C traps 1 cm^3 of air. Assume that its pressure remains unchanged during heating.

(a) Find the volume of this air when the meatball is heated to 100 °C. (2 marks)

(b) Sketch a graph to show how the volume of the air varies with its temperature. (2 marks)

*17 Misc A weather balloon contains 5 m^3 of helium at a pressure of 105 kPa and a temperature of 27 °C (Fig l).



Fig l

(a) Find the number of moles of gas inside the weather balloon. (2 marks)

(b) The balloon rises to a certain height so that the pressure and temperature of the gas inside become 80 kPa and 7 °C.

(i) Find the volume of the gas inside the balloon at that height. (2 marks)

(ii) Find the change in the internal energy of the gas. (2 marks)

(iii) Find the ratio of the r.m.s. speed of the helium gas molecules before and after the balloon rises in height. (2 marks)

*18 5.1 A cylinder contains an ideal gas at 20 °C and 100 kPa. The gas is trapped by a light piston with a radius of 5 cm resting at a height of 20 cm above the bottom of the cylinder (Fig m).

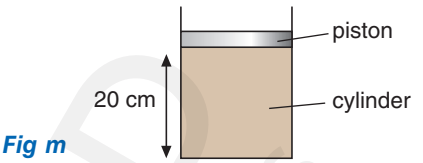


Fig m

(a) How many moles of gas molecules are there inside the cylinder? (2 marks)

(b) The gas is then compressed when a weight of 200 N is placed on the piston.

(i) Find the new pressure of the gas. (2 marks)

(ii) If the temperature of the gas increases to 21 °C during the compression, what is the new height of the piston? (2 marks)

*19 Misc A diver is 10 m below the water surface (Fig n), where the temperature (in kelvin) is 10% lower than that at the water surface and the pressure is much higher than the normal atmospheric pressure.



Fig n

(a) Explain, in terms of heat transfer processes, why the water temperature is higher at the surface than it is at 10 m below the surface. (4 marks)

(b) When a gas bubble rises to the water surface from 10 m below, it expands and the gas pressure inside is halved. Find the percentage change in the volume of the gas bubble. (2 marks)

(c) Hence, explain the danger that a diver may face if he does not regulate the air pressure in his lungs properly when ascending. (2 marks)

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Ext

★20 5.2 Based on the kinetic theory of an ideal gas, we have the equation $pV = \frac{1}{3}Nm\overline{c^2}$.

(a) Show how the total kinetic energy of the molecules is related to its number of moles and its absolute temperature. (2 marks)

(b) Find the total kinetic energy of 3.5 mol of ideal gas at 70 °C. (2 marks)

(c) State two conditions for a real gas to behave like an ideal gas. (2 marks)

★21 5.1 Daisy is going to study the p–T relationship of an ideal gas. She immerses a sealed flask filled with the ideal gas in a hot water bath (Fig o). She measures the pressure p of the gas at various temperatures T.

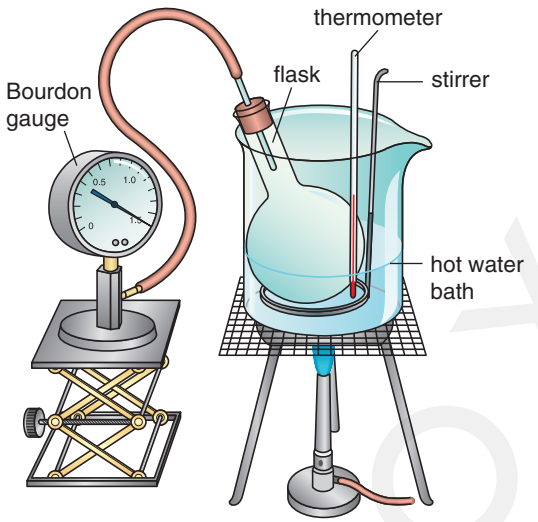


Fig o

(a) Point out all the mistakes in her set-up. (3 marks)

(b) Daisy corrects the mistakes in (a) and obtains the following data (Table a).

p / kPa	100	105	110	115	120
T / °C	20	35	50	65	80

Table a

Plot a graph of p against T. (2 marks)

(c) Using your graph in (b), find the pressure of the gas when the temperature is 320 K. (1 mark)

★22 5.1 Read the following passage about **the Magdeburg hemispheres** and answer the questions that follow.

Before the concept of atmospheric pressure was well established, there was a famous experiment that could demonstrate the existence and the power of atmospheric pressure.

In the experiment, two identical metal hemispherical shells were sealed together to form a single hollow sphere (Fig p). Once all the air was pumped out of the sphere via a valve, even two teams of eight horses moving in opposite directions failed to separate the two shells (Fig q) until the valve was opened to allow air to move back into the sphere. This famous experiment was performed in Magdeburg (a city in Germany). For this reason, the two shells are now commonly referred to as the Magdeburg hemispheres.



Fig p

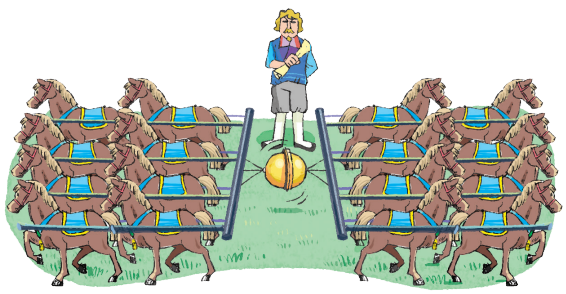


Fig q

(a) Why was it difficult to separate the two shells after the air was pumped out? (2 marks)

Ext

(b) Suppose the experiment was conducted with two cylindrical 'cups' rather than hemispherical shells. Assume the diameter of each cup was 50 cm. Calculate the minimum force F required to act on each cup in the direction shown in Figure r to separate them when all the air was pumped out. [Hints: pressure measures how concentrated a force is on a surface area.] (2 marks)

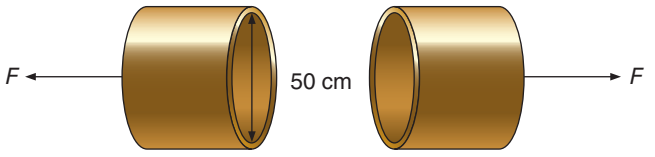


Fig r

★★23 5.1 An experiment is performed to study Charles' law (Fig s).

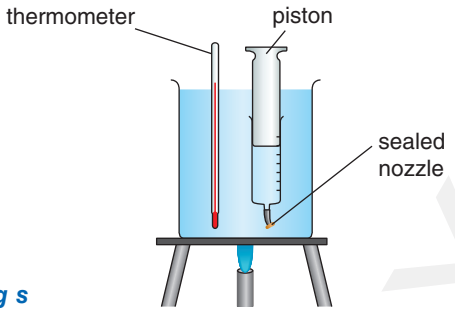


Fig s

(a) State one way to improve the accuracy of the experiment. (1 mark)

(b) Explain why it is not suitable to heat the gas inside the syringe directly using a Bunsen burner. (2 marks)

(c) The following table shows the volumes of the gas column V at various temperatures T (Table b).

V / cm^3	40	42	44	46	48
T / °C	20	35	47	64	76

Table b

(i) Plot a V–T graph with T in degrees Celsius. (2 marks)

(ii) Determine the absolute zero from the graph in (c)(i). Assume that the gas is an ideal gas. (1 mark)

(iii) The gas is kept at 100 kPa throughout the experiment. Find the number of moles of gas molecules. (3 marks)

Refer p.151–152

★★24 Misc Figure t shows a closed syringe containing an ideal gas. The piston has a negligible weight and it can move freely along the syringe. The gas is initially at room temperature (25 °C).

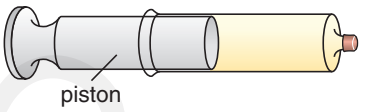


Fig t

(a) What is the pressure of the gas? (1 mark)

(b) The gas is heated slowly to a temperature of 80 °C. By considering the general gas law, explain how the piston would change its position. (3 marks)

(c) The gas is kept at 80 °C. The piston is pulled by an applied force to the left. Explain how the pressure of the gas would change using the kinetic theory of gases. (4 marks)

Refer p.169–170

★★25 5.1 A cylinder with a movable piston contains some ideal gas of volume V (Fig u). A pressure sensor is connected to the cylinder to measure the gas pressure p inside. Some weights are placed on the piston. The temperature of the gas is kept at 23 °C. A graph of p against $\frac{1}{V}$ is shown in Figure v.

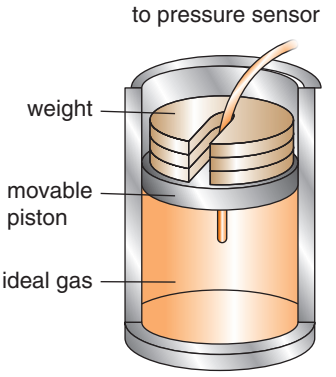


Fig u

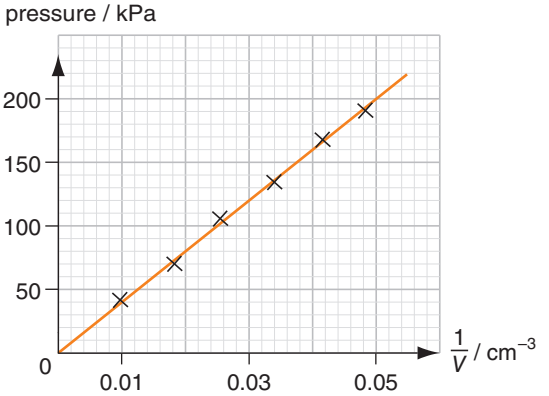


Fig v

Q27 Exam Report:

(a) Not many candidates were able to state the correct procedures to be done before taking a reading. They mentioned stirring the water but not stopping the heat.

(b) (i) Many candidates managed to work out the correct numerical answer but some of them used incorrect units.

(ii) The concept of absolute zero by experiment (L = 0) was not well interpreted by some candidates. Many of them just stated –273 °C as absolute zero without deducing the value from the data given.

Ext

(a) The number of weights on this piston is adjustable. Suggest the purpose of the weights in the set-up. (1 mark)

(b) (i) Show that the slope of the graph is equal to nRT, where n is the number of moles of gas molecules and T is the temperature of the gas. (1 mark)

(ii) Find the number of moles of the gas molecules. (3 marks)

(c) The experiment is then repeated at another constant temperature higher than 23 °C. Sketch a possible graph for it in Figure v. (1 mark)

Refer p.160

★★**26** 5.1 Peter sets up the following apparatus to study Charles' law (Fig w). A constant amount of ideal gas X is trapped under the paraffin oil inside a water bath.

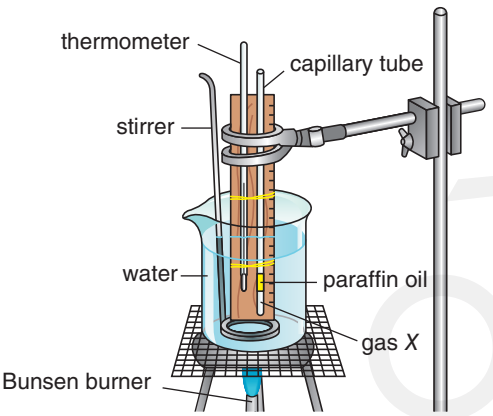


Fig w

(a) What is the required condition for Charles' law to hold in this set-up? (1 mark)

(b) Without using any extra apparatus, how can Peter ensure that the condition in your answer to (a) is satisfied? (1 mark)

(c) What is the pressure of gas X if the weight of the paraffin oil is negligible? (1 mark)

(d) Describe the procedure of the experiment. State clearly how the results can be obtained. (4 marks)

(e) Explain the potential danger of stirring the water using the thermometer rather than the stirrer. (1 mark)

Refer p.151–152

5.1 **27** **HKDSE 2013 Paper 1B Q2**

Figure x shows an air column trapped by a small mercury thread inside a uniform capillary tube. The set-up is heated by a water bath. The length of the air column L is measured at various temperature θ.

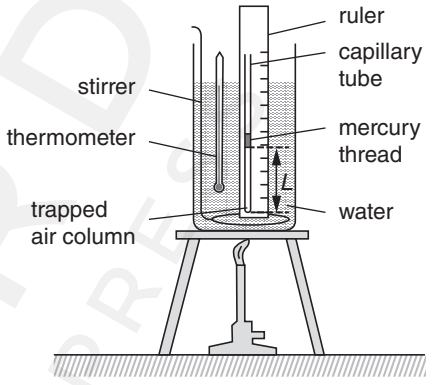


Fig x

Some of the results are tabulated below:

Temperature θ / °C	20		92
Length of the air column L / mm	64		80

Table c

(a) Describe the procedure(s) to be done before taking a reading in order to ensure that the trapped air reaches the same temperature as the water. (2 marks)

(b) Assume that length L increases linearly with temperature θ throughout.

(i) Estimate the length of the air column when the temperature indicated by the thermometer is 65 °C. (2 marks)

(ii) Find the 'absolute zero' as obtained from this experiment. (2 marks)

Misc **28** **HKDSE 2014 Paper 1B Q2**

See p.181 for the Exam Report.

Figure y shows a basketball connected to an air pump via a short rubber tubing. By pushing the piston inward, the pump can compress 120 cm^3 of air inside its barrel at atmospheric pressure and room temperature into the basketball for each stroke.

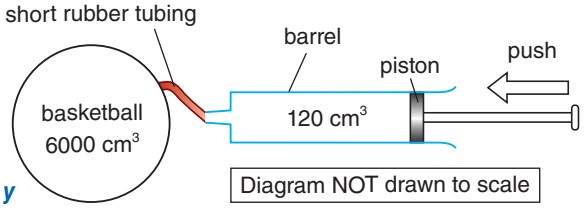


Fig y

Ext

Initially the volume of air inside the basketball is 6000 cm^3 and is at equilibrium with the atmospheric pressure 100 kPa. The basketball has to be pumped to a pressure of 156 kPa for an official match. Throughout the pumping process, the temperature of the basketball and the surroundings is assumed to be maintained at room temperature which is constant.

For each stroke, the valves of the pump (not shown in Figure y) allow the air in the barrel to be completely pumped into the basketball and prevent it from going back into the barrel when the piston is pulled outward.

(a) (i) Show that 3360 cm^3 of air, originally at atmospheric pressure, is required to be pumped into the basketball until its pressure is suitable for an official match. Assume that the volume of the basketball remains unchanged at 6000 cm^3. (3 marks)

(ii) Hence estimate the minimum number of strokes needed to pump the basketball to the required pressure. (1 mark)

(b) Use kinetic theory of an ideal gas to explain the increase of pressure inside the basketball when air is pumped into it. (2 marks)

5.1 **29** **HKDSE 2016 Paper 1B Q2**

See p.181 for the Exam Report.

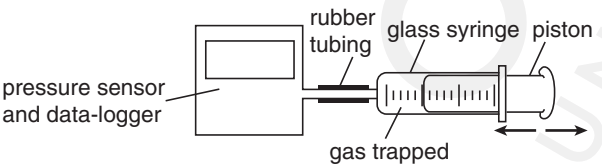


Fig z

Judy uses the set-up shown in Figure z to study the relationship between the pressure and volume of a fixed mass of gas at constant temperature. The volume V of the gas trapped is read directly from the syringe and the corresponding pressure p is measured by a data-logger via a pressure sensor.

(a) The initial volume and pressure of the gas are 6.0×10^{-5} m^3 and 1.0×10^5 Pa respectively at a room temperature of 25 °C. Estimate the number of gas molecules trapped in the syringe. (3 marks)

(b) The piston is then pushed in or pulled out to vary V and p such that several pairs of readings are recorded. Figure aa shows the graph of V against $\frac{1}{p}$ plotted.

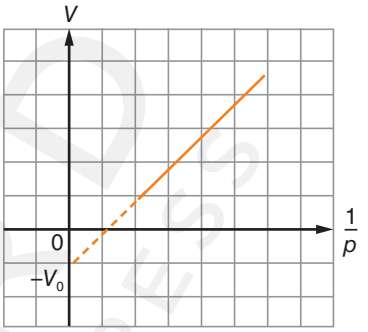


Fig aa

(i) State **one** experimental precaution for keeping the gas temperature constant. (1 mark)

(ii) The straight line graph does not pass through the origin but cuts the vertical axis at $-V_0$ instead. Suggest what V_0 stands for. (1 mark)

(iii) If the experiment is repeated at a higher room temperature using this set-up with the same mass of the same gas, sketch the expected graph in Figure aa. (2 marks)

5.1 **30** **HKDSE 2019 Paper 1B Q2**

See p.182 for the Exam Report.

A weather balloon of volume 0.52 m^3 is filled with helium gas of temperature 15 °C and pressure 100 kPa at ground level (Fig ab).



Fig ab

(a) Find the amount of helium gas (in mol) in the balloon. (2 marks)

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Ext

(b) The following graphs show the variation of air temperature T and atmospheric pressure p with height x above ground level (Fig ac and ad).

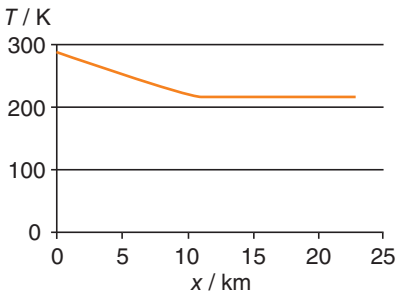


Fig ac

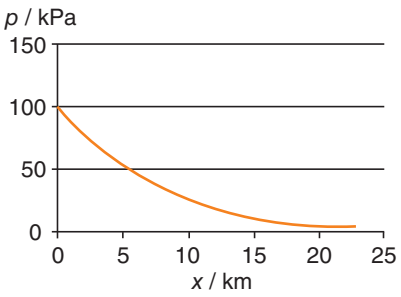


Fig ad

The weather balloon is released and rises to the upper atmosphere. Assume that the temperature and pressure of the helium gas in the balloon are the same as those of the air outside at any height x.

(i) A student believes that as the air temperature decreases in the first 10 km, the volume of the balloon decreases. Referring to the graphs above, explain qualitatively why this belief is not correct. (2 marks)

(ii) In fact the weather balloon keeps on expanding when it rises. The air temperature becomes steady at 216 K from a height of 12 km onwards. When the balloon rises further beyond 12 km and its volume reaches 8 m^3,

(1) estimate the gas pressure in the balloon; (2 marks)

(2) hence find the corresponding height reached by the balloon. The variation of atmospheric pressure p with height x (in km) is given by $p = p_0 e^{-kx}$, where p_0 is the atmospheric pressure at ground level and $k = 0.138\ km^{-1}$. (2 marks)

Video

→ This video demonstrates the Phone experiment.

You may search the app on the Internet.

physics phone experiment app

Phone experiment

Boyle's law (p.151)

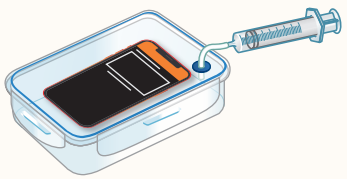
→ Worksheet is available on OUP web.

What to do

1 Use an air-tight food container with air vent on its lid. Measure and record the capacity of the container and the volume of the smartphone used. The smartphone must have a built-in pressure sensor.

2 Lubricate the piston of a syringe with water so that it can slide smoothly into the syringe. Remove all the air from the syringe.

3 Use plasticine to make the vent sealing ring fit the lid more firmly. Insert a short rubber tubing through the vent hole. Connect the other end of the tubing to the syringe. Seal all joints with plasticine. If the tubing is too thin to fit tightly into the hole, wrap adhesive tape around the tubing to make it thicker.



4 Slowly pull the piston. Record several sets of pressure in the container by using the app and the volume of the air in the syringe.

What you find

What is the relationship between the pressure and the volume of a gas?